

Online Appendix to Competition Creates Competition

A. Full analytical arguments

This appendix contains the complete arguments behind the results in the paper. I begin with the probability space, because every later claim about beliefs, deviations, and null sets rests on it, and then take the results in the order of the paper.

A.1. Probability space, strategies, and conditional information

The benchmark has a risk-neutral seller, an incumbent, a potential challenger, a strategic investor, competitive market makers, and exogenous noise demand. A known target standalone value is normalized to zero. The seller commits before trading to a binding second-price cash auction with reserve p . The incumbent has value $R \sim U[0, r]$, pays no incremental participation cost at the modeled stage, and learns its value before bidding. The challenger has value ℓ or h with equal probability, where $0 < p < \ell < r < h$. Neither bidder trades target equity. An outside investor observes challenger quality and chooses an order in $[-1, 1]$, pays $k|q|$, and earns $q(V_T - P) - k|q|$. Its order unit is not a controlling ownership stake. All payoffs and preparation costs use the same per-share normalization.

Noise has density $f(z) = e^{-|z|/b}/(2b)$, with $b > 1$ and $k > 0$. Market makers observe $X = q + Z$ and price $P(X) = \mathbb{E}[V_T | X]$. The challenger observes P but not X , then its independent preparation cost, equal to c_L with probability ρ and c_H otherwise, where $0 < \rho < 1$ and $0 \leq c_L < c_H$. Paying the cost reveals its value and permits bidding; declining means absence. Both bidders bid truthfully once informed. A bid meeting the reserve is admissible. I prescribe preparation at equality of gross expected profit and cost. The allocation, cash payment, and stock payoff are then realized. The equilibrium requires Bayesian pricing, optimal preparation at every reached price and cost, and global investor best responses against the fixed equilibrium price and preparation schedules, permitting mixed orders.

I write t_0 for expected target proceeds without challenger entry, t_H, t_L for proceeds conditional on entry and quality, and g_H, g_L for conditional gross challenger profits. Set $\Delta_T = t_H - t_L$, $B_r(\mu) = g_L + \mu(g_H - g_L)$, $m = (1 + e^{2/b})^{-1}$, and $M = 1 - m$. For the comparison $p < \ell < r_0 < r_1 < h$, the hypotheses of Proposition 2, referred to below as (A1) to (A3), are

$$\begin{aligned} 0 &\leq c_L < B_{r_1}(m), \\ B_{r_0}(1/2) &< c_H < B_{r_1}(M), \\ \Delta_T(r_0) &< k < (1 - 1/b)\rho m \Delta_T(r_1). \end{aligned} \tag{OA.1}$$

I give a probability-space construction so that every conditioning step and every deviation is explicit. Let $\Theta \in \{L, H\}$ have equal probabilities, with acquisition values $v_L = \ell$ and $v_H = h$. Independently draw

incumbent value R , preparation cost C , noise demand Z , and an auxiliary random variable U uniform on $[0, 1]$. A mixed trader strategy is a probability kernel σ_Θ on the Borel subsets of $[-1, 1]$. It can be implemented by a Borel quantile map $q_\Theta(U)$. Conditional randomization therefore contains no information about the other primitives beyond the trader's specified signal.

All spaces are standard Borel. Prices, entry policies, and payoffs are Borel functions. Because values and the order interval are bounded, all acquisition and trading payoffs are integrable. A regular conditional distribution of quality given price exists. Expectations conditional on a price atom use its complete preimage, not a pointwise inversion that presumes injectivity.

For the benchmark, define

$$a_\theta(x) = \int_{[-1,1]} f(x-q) \sigma_\theta(dq), \quad \nu(dx) = \frac{a_H(x) + a_L(x)}{2} dx, \quad \mu_X(x) = \frac{a_H(x)}{a_H(x) + a_L(x)}. \quad (\text{OA.2})$$

The density f is positive everywhere, bounded, continuous, and integrates to unity. Tonelli's theorem gives $\int a_\theta = 1$. Dominated convergence in q implies continuity of $a_\theta(x)$, and positivity gives a continuous posterior. The distribution of X under every pure or mixed unilateral order deviation is absolutely continuous with respect to Lebesgue measure. Moreover, ν is equivalent to Lebesgue measure because its density is strictly positive. Thus an equilibrium identity holding ν -almost surely also holds almost surely under every unilateral trading deviation. No deviation can exploit a different version of prices or beliefs on a null set.

Write $\mathcal{G} = \sigma(P(X))$ and $\mu_P = \mathbb{E}[\mathbf{1}\{\Theta = H\} \mid \mathcal{G}]$. Since $P(X)$ is measurable with respect to X ,

$$\mu_P = \mathbb{E}[\mu_X(X) \mid \mathcal{G}]. \quad (\text{OA.3})$$

For any $q, q' \in [-1, 1]$, the triangle inequality implies

$$||x - q| - |x - q'|| \leq |q - q'| \leq 2. \quad (\text{OA.4})$$

Exponentiating and integrating with respect to $\sigma_H(dq)\sigma_L(dq')$ gives $e^{-2/b}a_L \leq a_H \leq e^{2/b}a_L$, hence $m \leq \mu_X \leq M$. Equation (OA.3) preserves these bounds for μ_P . These conclusions do not impose pure trading, monotone orders, or an informative price.

The bidder's preparation rule is $\mathbf{1}\{C \leq B_r(\mu_P)\}$, with entry at equality. When $c_L < B_r(m)$, the cost-averaged rule is bounded below by ρ . Conditional on Θ and X , the variables R, C remain independent of the investor's action and retain their specified laws. This is why the auction-stage expectations can be used inside the pricing equation. It also explains why a unilateral order deviation shifts the density of X without changing the conditional auction formulas.

For the complementary-signal extension, I replace the investor's observation of Θ by T and add Y . The randomization U , noise Z , incumbent R , and cost C remain independent of (Θ, T, Y) ; T and Y are independent conditional on Θ . Sections A.7 and C.3 specify the resulting conditional laws rather than treating the buyer's private signal as public.

A.2. Auction implementation and Proposition 1

Conditional on the highest competing admissible bid, truthful bidding maximizes a private-value bidder's payoff in a second-price auction. A bid above value can turn a loss into an unprofitable win, and a bid below value can discard a profitable win. Conditional on winning, the bidder's own bid does not determine payment. The presence of a public price or information conveyed by entry changes beliefs about other bidders but not this pointwise dominance argument. I use truthful implementation to fix payoff-equivalent bid descriptions.

Without the challenger, the seller receives p if $R \geq p$. A high-value challenger wins almost surely and pays $\max(p, R)$ because h exceeds incumbent support. When the challenger has value ℓ , the winning value is $\max(R, \ell)$ and the runner-up is $\min(R, \ell)$, so seller payment is $\max(p, \min(R, \ell))$. The challenger obtains $(\ell - \max(p, R))_+$.

The uniform formulas follow by splitting at p and ℓ :

$$\begin{aligned} t_0 &= \frac{1}{r} \int_p^r p \, du, \\ t_H &= \frac{1}{r} \left[p^2 + \frac{r^2 - p^2}{2} \right], \\ t_L &= \frac{1}{r} \left[p^2 + \frac{\ell^2 - p^2}{2} + \ell(r - \ell) \right], \\ g_L &= \frac{1}{r} \left[p(\ell - p) + \frac{(\ell - p)^2}{2} \right], \quad g_H = h - t_H. \end{aligned} \tag{OA.5}$$

Evaluating these expressions gives $t_0 = p(1 - p/r)$, $t_H = r/2 + p^2/(2r)$, $t_L = \ell - (\ell^2 - p^2)/(2r)$, $g_H = h - t_H$, and $g_L = (\ell^2 - p^2)/(2r)$. Subtracting gives $\Delta_T = (r - \ell)^2/(2r)$. On $p < \ell < r < h$, the derivatives are $\Delta'_T = 1/2 - \ell^2/(2r^2) > 0$, $g'_H = -1/2 + p^2/(2r^2) < 0$, and $g'_L = -(\ell^2 - p^2)/(2r^2) < 0$. Thus $\partial_r B_r(\mu) < 0$ at every fixed posterior. In particular $g_H - g_L > 0$, since the high challenger has a strictly higher value in a mechanism whose allocation is monotone in its own value.

For a general continuous incumbent CDF F on $[0, \bar{r}]$, the payment difference is zero for $R \leq \ell$ and $R - \ell$ for $R > \ell$. By writing a positive part as an integral of indicators,

$$(R - \ell)_+ = \int_\ell^{\bar{r}} \mathbf{1}\{R > u\} \, du, \quad (\theta - \max(p, R))_+ = \int_p^\theta \mathbf{1}\{R < u\} \, du. \tag{OA.6}$$

Tonelli applies because the integrands are nonnegative and the domains bounded. CDF continuity removes the distinction between $R < u$ and $R \leq u$ in expectation; even without continuity the Lebesgue integral over u is unaffected by countably many atoms. Thus the two expressions are respectively $\int_\ell^{\bar{r}} (1 - F(u)) \, du$ and $\int_p^\theta F(u) \, du$, extending $F = 1$ above support. A first-order stochastic strengthening lowers F pointwise. Nonnegative integral differences prove weak ordering; a strictly positive integral difference proves strictness. Posterior-weighted challenger profits inherit the ordering. This completes Proposition 1.

A.3. Price sufficiency, residuals, and convolution regularity

For any candidate equilibrium with the low-cost entry floor, the independent cost distribution gives a Borel function $e(P) \geq \rho$. Conditional on $X = x$, expected target payoff is

$$P(x) = t_0 + e(P(x))\{t_L - t_0 + \Delta_T \mu_X(x)\}. \quad (\text{OA.7})$$

The function $g(P) = [P - t_0 - e(P)(t_L - t_0)]/[e(P)\Delta_T]$ is Borel on the realized price range because its denominator is strictly positive. It satisfies $\mu_X(X) = g(P(X))$ almost surely. Equation (OA.3) then gives $\mu_P = \mu_X$ almost surely. The entry function is consequently

$$e(\mu) = \rho + (1 - \rho)\mathbf{1}\{g_L + \mu(g_H - g_L) \geq c_H\}. \quad (\text{OA.8})$$

For construction, put $w_L = t_L - t_0 > 0$. For $\mu_2 > \mu_1$,

$$\begin{aligned} P(\mu_2) - P(\mu_1) &= e(\mu_2)\Delta_T(\mu_2 - \mu_1) + [e(\mu_2) - e(\mu_1)](w_L + \Delta_T \mu_1) \\ &\geq \rho\Delta_T(\mu_2 - \mu_1) > 0. \end{aligned} \quad (\text{OA.9})$$

Thus $P(\mu)$ is strictly increasing even with an entry jump. Its inverse on its image is measurable: it is the restriction of a generalized inverse defined through infima of upper level sets. Price values inside a skipped interval are not generated by any flow and need not receive artificial posterior mass. By contrast, a flat posterior tail maps into an actual price atom and is conditioned on as an atom.

Conditional on fundamental quality, expected terminal value under a unilateral trading deviation is still $t_0 + e(P(x))(t_\theta - t_0)$. Subtracting (OA.7) gives $A_H = e\Delta_T(1 - \mu_X)$ and $A_L = e\Delta_T\mu_X$. Both residuals are bounded in $[\rho m\Delta_T, \Delta_T]$. The signs exclude incorrectly signed orders. This derivation accounts for the endogenous preparation response before taking the investor's expectation over shifted noise.

Next I justify every differentiation used for continuous deviations. Let $f \in W^{1,1}(\mathbb{R})$, with positive density and $|f'| \leq Lf$ almost everywhere, and let A be bounded, measurable, and nonnegative. For sign $\epsilon \in \{-1, 1\}$, define

$$F_\epsilon(s) = \int_{\mathbb{R}} f(x - \epsilon s)A(x) dx. \quad (\text{OA.10})$$

For $s_1 < s_2$, the absolutely continuous representative of f satisfies

$$f(x - \epsilon s_2) - f(x - \epsilon s_1) = -\epsilon \int_{s_1}^{s_2} f'(x - \epsilon u) du. \quad (\text{OA.11})$$

The absolute integral of the right-hand side after multiplication by A is at most $\|A\|_\infty(s_2 - s_1)\|f'\|_1$. Fubini therefore permits interchanging the integrals. This proves absolute continuity of F_ϵ and

$$F'_\epsilon(s) = -\epsilon \int f'(x - \epsilon s)A(x) dx, \quad |F'_\epsilon(s)| \leq LF_\epsilon(s) \quad (\text{OA.12})$$

almost everywhere. Translation continuity of f' in L^1 also makes the displayed derivative continuous. One way to see the needed translation continuity without an additional smoothness assumption is to approximate f' in L^1 by compactly supported continuous functions, for which it holds uniformly; the approximation error is unchanged by translation. The difference quotient of (OA.11) then converges in L^1 to $-\epsilon f'(\cdot - \epsilon s)$, identifying the classical derivative everywhere. The almost-everywhere statement is already sufficient for the proof.

The Laplace density is in $W^{1,1}$: it is continuous at zero, absolutely continuous on every bounded interval,

and has integrable weak derivative $-\text{sgn}(x)f(x)/b$. Its kink contributes no atom to the first weak derivative because the density itself is continuous. Hence $L = 1/b$. This argument differentiates the translated density, not the discontinuous entry policy.

For $U(s) = sF(s) - ks$, absolute continuity and the product rule imply

$$U'(s) = F(s) + sF'(s) - k \geq (1 - sL)F(s) - k. \quad (\text{OA.13})$$

If the right-hand side is bounded below by a strictly positive constant on the allowed interval, integration shows that $U(s_2) > U(s_1)$ for every $s_2 > s_1$. This is global order optimality, not a local first-order condition.

A.4. Complete proof of Proposition 2

I first establish that each claimed strategy is necessary against every candidate equilibrium, and then construct an equilibrium with that strategy.

Weak incumbent. Proposition A.1 and (A1) give the entry floor under every conditional mixed order. Section A.3 gives the residual upper bound. For a correctly signed order of magnitude $s > 0$,

$$\mathbb{E}[\text{trading profit} \mid \Theta] \leq s\{\Delta_T(r_0) - k\} < 0. \quad (\text{OA.14})$$

An incorrectly signed order has strictly negative gross expected profit. Zero gives zero. Thus each type has the unique best response zero, irrespective of the candidate equilibrium schedules. A mixed distribution placing positive probability on a nonzero order cannot be optimal: its payoff is an average of strictly negative payoffs and zero, and is strictly negative if that probability is positive. There is no issue from arbitrarily small trades; a nonpositive integrable random payoff that is strictly negative on a positive-probability set has a negative expectation.

Under zero orders, $X = Z$ is independent of quality. Set $e = \rho$ and $P = t_0 + \rho[(t_H + t_L)/2 - t_0]$. Condition (A2) excludes high-cost preparation at the prior, and (A1) admits low-cost preparation. Competitive pricing is correct and the previous payoff calculation validates zero. This constructs the unique trading and entry outcome.

Strong incumbent. Again consider any candidate equilibrium, including mixed orders. Section A.3 gives $F_\theta(s) \geq \rho m \Delta_T(r_1)$, and (OA.13) with $L = 1/b$ gives

$$U'_\theta(s) \geq (1 - 1/b)\rho m \Delta_T(r_1) - k > 0. \quad (\text{OA.15})$$

Every smaller correctly signed magnitude is strictly inferior to the unit magnitude; incorrect signs are inferior to zero. Thus the only possible equilibrium orders are $q_H = 1, q_L = -1$.

Their conditional flow densities are $f(x - 1)$ and $f(x + 1)$. Bayes' rule gives the piecewise posterior

$$\mu(x) = \begin{cases} m, & x \leq -1, \\ (1 + e^{-2x/b})^{-1}, & -1 < x < 1, \\ M, & x \geq 1. \end{cases} \quad (\text{OA.16})$$

Use the preparation rule (OA.8) and pricing (OA.7). Equation (OA.9) makes the price invertible with respect to this posterior, so the buyer optimizes using exactly the information it observes. The global bounds

verify investor best responses, and the construction satisfies competitive pricing and truthful bidding. This proves existence as well as uniqueness of trading and entry outcomes. Price functions can differ at unrealized price values or flow null sets without creating a different economic outcome.

Entry and allocation. Since $g_H > g_L$, expensive preparation requires $\mu \geq \tau$. The strict inequalities $B_{r_1}(1/2) < B_{r_0}(1/2) < c_H < B_{r_1}(M)$ imply $\tau \in (1/2, M)$. Solving (OA.16) gives $x^* = b \logit(\tau)/2 \in (0, 1)$. The Laplace survival function is

$$\bar{F}_Z(z) = \begin{cases} 1 - \frac{1}{2}e^{z/b}, & z < 0, \\ \frac{1}{2}e^{-z/b}, & z \geq 0. \end{cases} \quad (\text{OA.17})$$

Hence $\alpha_H = \bar{F}_Z(x^* - 1)$ and $\alpha_L = \bar{F}_Z(x^* + 1)$ give the main-text formulas. Both are positive. State-specific entry is $e_\theta = \rho + (1 - \rho)\alpha_\theta$, total entry is $(e_H + e_L)/2$, and high-quality ownership is $e_H/2$ because $h > r$. Each comparison is strict.

Information ordering. A signal experiment is its pair of conditional distributions given quality. Applying a constant Markov kernel to the strong-economy price produces the weak-economy constant price experiment. Conversely, every state-independent kernel applied to a constant experiment has the same conditional law in both states. The strong experiment does not: on an interior price region its posterior is nonconstant, since the conditional flow likelihood ratio is nonconstant and the price reveals it. Thus no reverse garbling exists. Differences in mean payoffs across economies do not affect this comparison of signal experiments.

Nonemptiness. Section A.6 gives a direct primitive construction for every $h > \ell$. The specified benchmark is also strictly inside the condition region. All inequalities involve continuous primitive expressions away from the support boundaries, so their strict satisfaction persists on an open neighborhood.

A.5. Fixed experiments, threshold distinctions, and finite nonmonotonicity

For Proposition A.3, fix one joint distribution of (S, Θ, C) for both strengths. Let $\mu(S) = \Pr(H \mid S)$. For $r_1 > r_0$, the difference in entry indicators is

$$\mathbf{1}\{C \leq B_{r_0}(\mu(S))\} - \mathbf{1}\{C \leq B_{r_1}(\mu(S))\} = \mathbf{1}\{B_{r_1}(\mu(S)) < C \leq B_{r_0}(\mu(S))\}. \quad (\text{OA.18})$$

Taking expectations proves weak decrease and the exact strictness condition, without requiring an atomless cost distribution. This comparison holds within a fixed full-order regime because both conditional flow laws are unchanged when only r changes. It need not hold across different informative order profiles.

For Proposition A.4, define the maintained domain

$$\mathcal{D} = \{r \in (\ell, h) : c_L < B_r(m), B_r(1/2) < c_H\}. \quad (\text{OA.19})$$

Under pooling, the public posterior is $1/2$ and entry is ρ . A unilateral correctly signed order sees a constant residual $\rho\Delta_T/2$ and earns $s(\rho\Delta_T/2 - k)$. Thus pooling is an equilibrium exactly when this coefficient is nonpositive. Equality supports pooling although the trader is indifferent among correctly signed sizes against that constant candidate schedule; it does not make every alternative profile an equilibrium with recomputed prices.

Solving $\Delta_T(r) = d$ gives $(r - \ell)^2 = 2rd$, whose roots are $\ell + d \pm \sqrt{d^2 + 2\ell d}$. The plus root is the one

above ℓ . Writing $\mathbf{r}(d) = \ell + d + \sqrt{d^2 + 2\ell d}$ gives the exact pooling-existence boundary $r_N = \mathbf{r}(2k/\rho)$, the sufficient pooling-uniqueness boundary $\mathbf{r}(k)$, and the sufficient full-order uniqueness boundary $r_U = \mathbf{r}(k/[(1 - 1/b)\rho m])$. These boundaries are applied only within $\mathcal{D}$. A sufficient global bound does not identify the first existence of an informative equilibrium.

At $r > r_U$ in $\mathcal{D}$, the global derivative bound forces full orders. Expensive entry is feasible only if the attainable posterior reaches the required τ . The uniform mixed-strategy posterior bound makes $B_r(M) < c_H$ sufficient to exclude expensive entry in every equilibrium, regardless of the orders. The relevant equality is

$$Mr^2 - 2(Mh - c_H)r - [(1 - M)\ell^2 - p^2] = 0. \quad (\text{OA.20})$$

Because $B_r(M)$ is strictly decreasing on $r > \ell > p$, there is at most one root in the economic domain. The larger quadratic root gives that root when it lies in the domain; if both algebraic roots are positive, the other cannot lie in this strictly decreasing domain as a second crossing. Numerical specifications must verify the discriminant, the root residual, and the support and floor conditions rather than treating the formula as globally applicable.

For Laplace noise, the posterior M is attained on $x \geq 1$. At equality $B_r(M) = c_H$, entry-at-indifference admits the costly buyer on that complete tail. As $\tau \uparrow M$, the flow threshold approaches unity and

$$\alpha_H \rightarrow \frac{1}{2}, \quad \alpha_L \rightarrow \frac{1}{2}e^{-2/b}, \quad E \rightarrow \rho + \frac{1 - \rho}{4}(1 + e^{-2/b}). \quad (\text{OA.21})$$

For any strict infeasibility beyond the crossing, costly entry is zero. An alternative tie rule changes the value at the equality itself, not the strict comparisons on either side. With logistic noise the upper posterior is approached only as $x \rightarrow \infty$; its survival probability tends to zero, and entry converges continuously to ρ .

For Proposition 2(iii), impose (A1) to (A3) at r_0, r_1 , the low-cost floor at r_2 , the strong global derivative bound at r_2 , and $B_{r_2}(M) < c_H$. Parts (i) and (ii) of Proposition 2 give the first two unique outcomes; the bound and infeasibility condition give unique full orders but entry ρ at r_2 . This proves a finite rise and fall without a selection or uniqueness assertion for strengths between them.

A useful full-profile existence test is sharper than the uniform uniqueness bound. At a fixed full-order candidate,

$$J = F_H(1) = F_L(1) = \Delta_T \int e(x) \frac{f(x-1)f(x+1)}{f(x-1) + f(x+1)} dx. \quad (\text{OA.22})$$

The equality follows by substituting Bayes' rule into both residuals. The pointwise density-ratio bound implies $F_\theta(s) \geq e^{-(1-s)/b}J$. Since $(1 - s/b)e^{-(1-s)/b}$ is decreasing on $[0, 1]$, $k < (1 - 1/b)J$ makes every correctly signed marginal profit positive. It verifies existence of that profile, not uniqueness over other candidate schedules. The distinction is preserved in the numerical classifications.

A.6. Smooth noise, atomless costs, and primitive nonemptiness

For logistic noise,

$$f(z) = \frac{e^{-z/b}}{b(1 + e^{-z/b})^2}, \quad (\log f)'(z) = -\frac{1}{b} \tanh\left(\frac{z}{2b}\right). \quad (\text{OA.23})$$

It is positive and smooth, and $|f'| \leq f/b$ implies $\|f'\|_1 \leq 1/b$. Therefore it belongs to $W^{1,1}$ and all posterior and global trading bounds apply. Under full orders the posterior log odds equal

$$\log \frac{\mu(x)}{1 - \mu(x)} = 2 \log \cosh \left(\frac{x+1}{2b} \right) - 2 \log \cosh \left(\frac{x-1}{2b} \right). \quad (\text{OA.24})$$

Their derivative equals

$$\frac{1}{b} \left[\tanh \left(\frac{x+1}{2b} \right) - \tanh \left(\frac{x-1}{2b} \right) \right] > 0. \quad (\text{OA.25})$$

The limits are $-2/b$ and $2/b$, giving the open posterior range (m, M) . Every strictly interior threshold is crossed with positive probability. To derive its inverse, set $y = e^{x/b}$, $A = e^{1/b}$, and $w = \sqrt{\tau/(1-\tau)}$. The square root of the likelihood ratio equals $(Ay + 1)/(y + A)$. Solving $w = (Ay + 1)/(y + A)$ gives $y = (Aw - 1)/(A - w) > 0$. Thus $x_{\log}^* = b \log[(Aw - 1)/(A - w)]$. Its conditional survival probabilities are $\alpha_H = [1 + e^{(x_{\log}^* - 1)/b}]^{-1}$ and $\alpha_L = [1 + e^{(x_{\log}^* + 1)/b}]^{-1}$. These formulas and $E = \rho + (1 - \rho)(\alpha_H + \alpha_L)/2$ supply the complete entry comparison. The trade and information-ordering arguments complete Proposition A.5.

For Proposition A.6, let $H_C = \rho H_L + (1 - \rho)H_H$, with atomless component laws supported within $[c_L - \varepsilon_C, c_L + \varepsilon_C]$ and $[c_H - \varepsilon_C, c_H + \varepsilon_C]$. Retain (A3) and impose $c_L - \varepsilon_C \geq 0$, $c_L + \varepsilon_C < B_{r_1}(m)$, and

$$B_{r_0}(1/2) < c_H - \varepsilon_C < c_H + \varepsilon_C < B_{r_1}(M). \quad (\text{OA.26})$$

Use either the Laplace or logistic noise law. The entry rule is $e(\mu) = H_C(B_r(\mu))$. It is continuous and nondecreasing, and $e \geq \rho$ under the low-cost support restriction. Equation (OA.9) remains strictly positive even if the component densities are discontinuous or vanish on subintervals. In the weak economy, $e(1/2) = \rho$. In the strong economy, a posterior sufficiently close to M satisfies $B_r(\mu) > c_H + \varepsilon_C$; an event with positive probability then gives $e = 1$. All global trading comparisons are unchanged. This proof requires an atomless distribution, not a differentiable density. Differentiability is imposed separately when differentiating a seller objective.

For primitive nonemptiness, fix $h > \ell > p > 0$, $b > 1$, and $0 < \rho < 1$. At the continuous limit $r = \ell$,

$$g_H - g_L = h - \ell > 0, \quad B_\ell(M) - B_\ell(1/2) = (M - 1/2)(h - \ell) > 0. \quad (\text{OA.27})$$

Choose $r_1 \in (\ell, h)$ close enough to ℓ that $B_{r_1}(M) > B_\ell(1/2)$. Because $B_r(1/2)$ decreases in r , every $r_0 \in (\ell, r_1)$ then satisfies $B_{r_0}(1/2) < B_{r_1}(M)$. Choose r_0 sufficiently near ℓ that $\Delta_T(r_0) < (1 - 1/b)\rho m \Delta_T(r_1)$. Select k strictly inside this interval and c_H strictly between the two profit bounds. The quantity $B_{r_1}(m)$ is positive, so choose c_L strictly between zero and that bound. The constructed $c_L < c_H$ follows because $B_{r_1}(m) < B_{r_0}(1/2)$, using $m < 1/2$ and the monotonicity in both arguments. All inequalities have strict slack and persist by continuity. This is a mathematical nonemptiness construction, not a statement about empirical effect size at every value ratio.

Two further stability results are recorded for completeness.

Lemma OA.1 (analytical). *Replace the trading cost by $K(s) = ks + \gamma_q s^2/2$, $\gamma_q \geq 0$. Under (A1) and (A2), $\Delta_T(r_0) < k$, and $k + \gamma_q < (1 - 1/b)\rho m \Delta_T(r_1)$, the conclusions of Proposition 2 remain valid.*

Indeed $K(s) \geq ks$ preserves the weak-economy exclusion. At high strength, subtracting $K'(s) = k + \gamma_q s$

from the gross marginal profit leaves a strictly positive bound. The price construction, threshold, and information comparison are unchanged. This perturbation retains the position bound; it does not solve an unbounded-order game.

Lemma OA.2 (analytical). *For prior $\pi \in (0, 1)$, let $\mu_- = \text{logistic}(\text{logit } \pi - 2/b)$, $\mu_+ = \text{logistic}(\text{logit } \pi + 2/b)$, and $\omega = \min\{\mu_-, 1 - \mu_+\}$. Replace (A1)–(A3) by $c_L < B_{r_1}(\mu_-)$, $B_{r_0}(\pi) < c_H < B_{r_1}(\mu_+)$, and $\Delta_T(r_0) < k < (1 - 1/b)\rho\omega\Delta_T(r_1)$. Then the benchmark entry reversal and unique trading outcomes hold.*

Prior odds multiply the order-flow likelihood ratio. The uniform residual lower bound is therefore $\rho\omega\Delta_T$, while the upper bound is still Δ_T . Every inference and global deviation argument follows with these bounds. Under full Laplace orders, $x^* = b[\text{logit } \tau - \text{logit } \pi]/2$ lies in $(0, 1)$, and total entry is $\rho + (1 - \rho)[\pi\alpha_H + (1 - \pi)\alpha_L] > \rho$. Strict feasibility at the equal prior persists for nearby priors. Neither supplemental lemma is required to generate a reported numerical value in the present manuscript.

A.7. Full proof and probability formulas for Proposition A.7

The auction, value supports, order interval, noise law, and preparation-cost distribution are those in A.1. Only the information observations change. Let $a, d \in (1/2, 1)$ denote investor and buyer signal accuracy. Define

$$\begin{aligned} \mu_- &= (1 - a) + (2a - 1)m, & \mu_+ &= (1 - a) + (2a - 1)M, \\ \phi_+(\mu) &= \frac{d\mu}{d\mu + (1 - d)(1 - \mu)}, & \phi_-(\mu) &= \frac{(1 - d)\mu}{(1 - d)\mu + d(1 - \mu)}, \\ w_H(r) &= t_H(r) - t_0(r), & w_L(r) &= t_L(r) - t_0(r). \end{aligned} \tag{OA.28}$$

The complete sufficient conditions for Proposition A.7 are

$$\begin{aligned} c_L &< B_{r_1}(\phi_-(\mu_-)), \\ B_{r_0}(d) &< c_H < B_{r_1}(\phi_+(\mu_+)), \\ (2a - 1)\{\Delta_T(r_0) + (1 - \rho)(2d - 1)w_H(r_0)\} &< k \\ &< (1 - 1/b)m(2a - 1)\rho\Delta_T(r_1). \end{aligned} \tag{OA.29}$$

I prove unique zero orders and entry ρ at r_0 , unique maximum correctly signed signal-contingent orders at r_1 , and strictly higher entry at r_1 . The result allows $d > a$ and arbitrary mixed orders conditional on the investor's signal.

I use T for the trader's signal, Y for the buyer's private signal, and Θ for fundamental quality. Conditional on H , the probabilities of $T = +$ and $Y = +$ are a and d ; conditional on L they are $1 - a$ and $1 - d$. Conditional independence gives the joint signal law as the product of these probabilities within each fundamental state. Equal fundamental priors imply equal marginal probabilities for each trader signal.

For arbitrary signal-contingent mixed orders, let $a_+(x) = \int f(x - q)d\sigma_+(q)$ and $a_-(x) = \int f(x - q)d\sigma_-(q)$. Then

$$\lambda_X = \frac{a_+}{a_+ + a_-} \in [m, M], \quad \mu_X = \frac{aa_+ + (1 - a)a_-}{a_+ + a_-} = (1 - a) + (2a - 1)\lambda_X. \tag{OA.30}$$

Since X depends on quality only through T , Θ and X are conditionally independent given T . Also Y is independent of X conditional on Θ . The latter remains true after applying a measurable price function, and

implies

$$\Pr(H \mid P, Y = y) = \phi_y(\mu_P). \quad (\text{OA.31})$$

The maps ϕ_+ , ϕ_- are continuous and strictly increasing on $(0, 1)$; the favorable private signal gives the larger posterior. The public posterior based on price inherits $[\mu_-, \mu_+]$ by conditional expectation. The lowest possible joint posterior is consequently $\phi_-(\mu_-)$. The low-cost restriction in (OA.29) and declining gross profits in r make low-cost preparation optimal in both economies under any candidate equilibrium.

Define high-cost indicators $I_y(P) = \mathbf{1}\{B_r(\phi_y(\mu_P)) \geq c_H\}$. They obey $I_+ \geq I_-$. Since Y 's law conditional on quality does not change with X or the trader's signal, true-state entry conditional on P is

$$\begin{aligned} e_H(P) &= \rho + (1 - \rho)\{dI_+(P) + (1 - d)I_-(P)\}, \\ e_L(P) &= \rho + (1 - \rho)\{(1 - d)I_+(P) + dI_-(P)\}. \end{aligned} \quad (\text{OA.32})$$

In particular,

$$e_H - e_L = (1 - \rho)(2d - 1)(I_+ - I_-), \quad e_H \geq e_L \geq \rho. \quad (\text{OA.33})$$

Let $D = e_H w_H - e_L w_L$. As $w_H - w_L = \Delta_T$ and $w_L > 0$,

$$D = e_L \Delta_T + (e_H - e_L)w_H \in [\rho \Delta_T, \Delta_T + (1 - \rho)(2d - 1)w_H]. \quad (\text{OA.34})$$

Conditional pricing is $P = t_0 + e_L(P)w_L + \mu_X D(P)$. Both entry probabilities are functions of price, even before proving sufficiency. The positive coefficient $D(P)$ therefore makes μ_X measurable with respect to price by the explicit inversion. The tower property gives $\mu_P = \mu_X$ and hence also reveals λ_X through the affine inverse in (OA.30).

For construction, set the entry indicators using $\phi_y(\mu)$ and write

$$P(\mu) = t_0 + \mu e_H(\mu)w_H + (1 - \mu)e_L(\mu)w_L. \quad (\text{OA.35})$$

Both entry probabilities are nondecreasing in μ . For $\mu_2 > \mu_1$, decompose its change into a belief change at the old entry probabilities and an entry change evaluated at the new belief:

$$\begin{aligned} P(\mu_2) - P(\mu_1) &= (\mu_2 - \mu_1)D(\mu_1) + \mu_2[e_H(\mu_2) - e_H(\mu_1)]w_H \\ &\quad + (1 - \mu_2)[e_L(\mu_2) - e_L(\mu_1)]w_L > 0. \end{aligned} \quad (\text{OA.36})$$

Thus the same measurable-inverse construction works even when the entry indicators jump simultaneously or one never changes. This argument does not rely on differentiating an indicator.

Conditional on $T = +$, the trader assigns quality probability a even after its own randomized order and the resulting flow are specified. Conditional on $T = -$, it assigns $1 - a$. This remains valid under unilateral orders because the deviation introduces no additional quality information conditional on the signal. Its expected terminal target payoff at a flow x is therefore $t_0 + e_L(P(x))w_L + aD(P(x))$ for $T = +$, and the corresponding expression with $1 - a$ for $T = -$. Subtracting (OA.35) and using (OA.30) gives

$$A_+ = (2a - 1)(1 - \lambda_X)D, \quad A_- = (2a - 1)\lambda_X D. \quad (\text{OA.37})$$

The residuals have strict signs and obey

$$m(2a - 1)\rho\Delta_T \leq A_+, A_- \leq (2a - 1)\{\Delta_T + (1 - \rho)(2d - 1)w_H\} =: \bar{A}. \quad (\text{OA.38})$$

The weak restriction in (OA.29) makes any correctly signed order of magnitude $s > 0$ earn at most $s(\bar{A} - k) < 0$. Hence zero is necessary under all candidate equilibria. With zero orders the public posterior is the prior, the buyer's favorable posterior is d , and the expensive-entry exclusion in (OA.29) gives entry ρ . Constant competitive pricing constructs the weak equilibrium.

At high strength, the convolution regularity applies to either bounded signal residual and gives $U'(s) \geq (1 - 1/b)m(2a - 1)\rho\Delta_T - k > 0$. Thus the necessary and unique trading profile is full correctly signed orders by trader signal. Bayes' rule, (OA.35), and optimal preparation construct the equilibrium. The strict high-cost upper inequality ensures that $Y = +$ and public beliefs sufficiently near μ_+ induce preparation. Full Laplace orders attain those beliefs on a positive-probability event; the conditional probability of $Y = +$ is positive. Total entry strictly exceeds ρ . The proof allows $d > a$ throughout.

For numerical implementation with either private signal, let $\tau = (c_H - g_L)/(g_H - g_L)$ and solve $\phi_y(\mu) = \tau$:

$$\mu_{\text{req},+} = \frac{\tau(1 - d)}{d(1 - \tau) + \tau(1 - d)}, \quad \mu_{\text{req},-} = \frac{\tau d}{(1 - d)(1 - \tau) + \tau d}, \quad \lambda_{\text{req},y} = \frac{\mu_{\text{req},y} - (1 - a)}{2a - 1}. \quad (\text{OA.39})$$

These formulas presume an interior fundamental threshold. Thresholds outside $[0, 1]$ imply always or never entry and are handled directly by the profit comparison. For a requirement strictly between m and M , full Laplace orders give $x_y^* = b \logit(\lambda_{\text{req},y})/2$. Requirements below the lower bound or above the upper bound imply always or never entry; equality at a plateau follows the tie rule. The buyer's unfavorable signal need not be assumed to preclude high-cost entry in a parameter sweep.

The fundamental-conditioned flow densities are

$$f_H^X(x) = af(x - 1) + (1 - a)f(x + 1), \quad f_L^X(x) = (1 - a)f(x - 1) + af(x + 1). \quad (\text{OA.40})$$

Let $Q_{\theta,y} = \Pr(X \geq x_y^* \mid \Theta = \theta)$, with the always/never cases evaluated accordingly. Conditional entry is

$$\begin{aligned} \bar{e}_H &= \rho + (1 - \rho)[dQ_{H,+} + (1 - d)Q_{H,-}], \\ \bar{e}_L &= \rho + (1 - \rho)[(1 - d)Q_{L,+} + dQ_{L,-}]. \end{aligned} \quad (\text{OA.41})$$

Thus $E = (\bar{e}_H + \bar{e}_L)/2$, $O_H = \bar{e}_H/2$, and $\mathcal{R}_T = t_0 + (\bar{e}_H w_H + \bar{e}_L w_L)/2$. Independently integrating the joint law of (Θ, T, Y, Z, C) must reproduce these expressions. Using the marginal law of Y instead of its conditional law given quality would give the wrong entry and pricing objects.

A.8. Verifiable-value bargaining

The bargaining comparison uses a different acquisition institution, specified here. After diligence, all buyer values are verifiable. The seller can implement a sale to the runner-up at that buyer's value, with acceptance at zero buyer surplus; therefore the disagreement payoff in negotiating with the best buyer is the next-best

value z . The best buyer has value $V \geq z$ and disagreement payoff zero. For $0 < \eta < 1$, its Nash transfer solves

$$\max_{P \in [z, V]} (P - z)^\eta (V - P)^{1-\eta}. \quad (\text{OA.42})$$

For $V > z$, strict concavity of the log objective gives the unique transfer $P = (1 - \eta)z + \eta V$. When $V = z$, feasible transfers collapse to that common value. At $\eta = 0$ I use the continuous limiting transfer z . With no challenger, the same rule has fallback zero and pays the seller ηR . Thus no-entry proceeds are specified by the institution as well as entry proceeds.

The highest-value buyer receives the target. For $R \leq \ell$, the high and low target payments are $(1 - \eta)R + \eta h$ and $(1 - \eta)R + \eta \ell$. For $R > \ell$, they are $(1 - \eta)R + \eta h$ and $(1 - \eta)\ell + \eta R$. Their difference is

$$\eta(h - \ell) + (1 - 2\eta)(R - \ell)_+. \quad (\text{OA.43})$$

A challenger receives $(1 - \eta)(\theta - R)$ when $\theta \geq R$ and zero otherwise. Taking expectations gives Proposition A.8 and applying FOSD yields its sign comparisons. For a uniform incumbent,

$$\Delta_\eta = \eta(h - \ell) + (1 - 2\eta)\frac{(r - \ell)^2}{2r}, \quad G_{H,\eta} = (1 - \eta)(h - r/2), \quad G_{L,\eta} = (1 - \eta)\frac{\ell^2}{2r}. \quad (\text{OA.44})$$

The transfer is always feasible between fallback and winning value. For $\eta > 1/2$ the competition derivative of the spread is negative; the spread itself remains positive because the high challenger has the higher value. The endpoint $\eta = 1$ extinguishes buyer rents and is not used to infer a positive-cost entry equilibrium. The comparison does not solve a private-value bargaining or first-price auction game; verifiability and fallback enforceability are part of the stated institution.

A.9. Matched welfare and the external-dividend diagnostic

I couple the feedback and price-hidden economies at $r = r_1$ using the same (Θ, R, C, Z) and full informed orders. The strong trading bound applies in the price-hidden economy with entry probability ρ , so this coupling uses equilibrium behavior in both environments. The high-cost type remains out when price information is unavailable because $B_{r_1}(1/2) < c_H$; all low-cost realizations enter in both environments.

Without challenger entry, allocation value is $R\mathbf{1}\{R \geq p\}$. With entry it is $\max(R, \theta)$ because $\theta > p$. Splitting at $R = p$ and $R = \theta$ gives

$$\max(R, \theta) - R\mathbf{1}\{R \geq p\} = (\theta - \max(p, R))_+ + p\mathbf{1}\{R < p\}. \quad (\text{OA.45})$$

If $R < p$, both sides equal θ ; if $R \geq p$, both sides equal $(\theta - R)_+$. Conditional on the signal and preparation cost, R retains its independent uniform law, so incremental expected allocation value net of cost is

$$B_r(\mu_X) - C + p \Pr(R < p) = B_r(\mu_X) - C + \frac{p^2}{r}. \quad (\text{OA.46})$$

Additional entry occurs only when its first two terms sum to a nonnegative value. The final term is strictly positive. For atomic costs, integrate against the marginal full-order flow density $g(x) = [f(x-1) + f(x+1)]/2$ to obtain

$$\Delta \mathcal{W} = (1 - \rho) \int g(x) \left[B_r(\mu_X(x)) - c_H + \frac{p^2}{r} \right] \mathbf{1}\{B_r(\mu_X(x)) \geq c_H\} dx > 0. \quad (\text{OA.47})$$

For an atomless high-cost component H_H , replace the integrand by

$$(1 - \rho) \int \left[B_r(\mu_X(x)) - c + \frac{p^2}{r} \right] \mathbf{1}\{c \leq B_r(\mu_X(x))\} H_H(dc). \quad (\text{OA.48})$$

The strict-support conditions ensure positive additional entry. Fubini is valid because values and the specified cost supports are bounded. The realized trading cost is the same in both environments; any resource interpretation of that cost therefore cancels. Transfers between bidders, shareholders, market makers, and noise traders are not additional allocation surplus.

Target proceeds increase by averaging $t_\theta - t_0 > 0$ over each state's additional-entry probability. For atomic costs the difference is $(1 - \rho)\{\alpha_H(t_H - t_0) + \alpha_L(t_L - t_0)\}/2 > 0$; for atomless costs integrate the corresponding state-specific tail probabilities over the high-cost component. As an independent numerical check, compute total expected allocation minus the full expected preparation cost in each environment:

$$\mathcal{W} = \frac{r^2 - p^2}{2r} + \frac{1}{2} \sum_{\theta} \bar{e}_{\theta} \left(g_{\theta} + \frac{p^2}{r} \right) - \mathbb{E}[C \mathbf{1}\{\text{entry}\}], \quad (\text{OA.49})$$

valid in the benchmark support region. Subtracting the price-hidden value must agree with (OA.47) or (OA.48).

For the matched-level diagnostic, attach a deterministic payoff $D_0 = \mathcal{R}_T^{\text{feedback}} - \mathcal{R}_T^{\text{hidden}}$ to the traded claim in the price-hidden economy. Its competitive price shifts by D_0 and $V_T - P$ is unchanged. The buyer's acquisition profits and information set do not change. This matches mean prices without manufacturing information. The external payoff is not counted as a real surplus gain and is not an available reserve or acquisition payment.

A.10. Continuous acquisition values and sale-design regularity

For realized challenger value v , define pointwise seller revenue

$$T(R, v, p) = \begin{cases} p \mathbf{1}\{R \geq p\}, & v < p, \\ \max\{p, \min(R, v)\}, & v \geq p, \end{cases} \quad G(R, v, p) = (v - \max(p, R))_+. \quad (\text{OA.50})$$

These formulas include no sale when both values fail the reserve. They are bounded, Borel, and can be integrated first over R and then over any conditional challenger distribution. The latter order is convenient when a reserve cuts through a value band. At a value atom equal to the reserve, the bid meets the reserve even when its acquisition rent is zero; the cost of preparation was paid before the realized value was learned.

For uniform R and a fixed v , a useful complete kernel is

$$\begin{aligned}
v < p : \quad & t_v = t_0, \quad g_v = 0, \\
p \leq v < r : \quad & t_v = v - \frac{v^2 - p^2}{2r}, \quad g_v = \frac{(v - p)^2}{2r} + \frac{p(v - p)}{r}, \\
p \leq r \leq v : \quad & t_v = \frac{r}{2} + \frac{p^2}{2r}, \quad g_v = v - t_v, \\
r < p \leq v : \quad & t_v = p, \quad g_v = v - p,
\end{aligned} \tag{OA.51}$$

with $t_0 = p(1 - p/r)$ for $p \leq r$ and $t_0 = 0$ for $p > r$. The third line presumes $p \leq r$; equalities agree across the applicable formulas. In the second line $g_v = (v^2 - p^2)/(2r)$. Direct integration of (OA.50), rather than the shortened kernels alone, is the independent numerical oracle.

For the atomless-value diagnostic, a binary class S is equally likely. Conditional on $S = L$, V is uniform on $[\ell - \varepsilon_V, \ell + \varepsilon_V]$; conditional on $S = H$, it is uniform on $[h - \varepsilon_V, h + \varepsilon_V]$. The investor observes S , not the exact V . Preparation reveals V . The independent incumbent remains uniform. Conditional class values have equal width but different means. This is a binary-information economy with atomless acquisition values, not a perfectly informed trader observing a continuous value.

When $p < \ell - \varepsilon_V < \ell + \varepsilon_V < r < h - \varepsilon_V$,

$$\begin{aligned}
t_L &= \ell - \frac{\ell^2 + \varepsilon_V^2/3 - p^2}{2r}, \quad g_L = \frac{\ell^2 + \varepsilon_V^2/3 - p^2}{2r}, \\
t_H &= \frac{r}{2} + \frac{p^2}{2r}, \quad g_H = h - t_H, \quad \Delta_T = \frac{(r - \ell)^2}{2r} + \frac{\varepsilon_V^2}{6r}.
\end{aligned} \tag{OA.52}$$

When the reserve is above the entire low band but below r , $t_L = t_0$, $g_L = 0$, and the high-class formulas are unchanged. Conditional on class and flow, exact V retains its within-class distribution, so the benchmark price and residual proof applies to these class-averaged payoffs whenever $\Delta_T > 0$ and a positive entry floor holds. In this region $t_L = t_0$ is permitted: the price construction remains strictly increasing on the bounded posterior interval because $\Delta_T \mu > 0$.

For a general reserve, the seller's continuation includes the public price posterior, not necessarily the raw order-flow posterior. If entry vanishes, the inversion denominator vanishes. A replication must condition on any resulting price pool. If $\Delta_T = 0$ or all entry is impossible, the informative-price argument cannot be imported. Section C.6 therefore requires direct pricing, posterior, and entry verification at such reserves.

For the local decomposition in the paper, impose a neighborhood I contained strictly in $0 < p < \ell$, a pure strategy branch $q_\theta(p)$ continuously differentiable on I , and an atomless preparation-cost CDF H_C continuously differentiable on the compact profit range. Suppose $\mu_\theta(z; p)$ is differentiable in p for almost every z and there is an integrable function $J(z)$ dominating

$$f(z) \left| h_C(B_{p,r}(\mu_\theta)) \left[-p/r + (g_H - g_L) \partial_p \mu_\theta \right] \right| \tag{OA.53}$$

uniformly on a compact neighborhood of the reserve. These assumptions are sufficient, not an assertion that every continuation has them. For positive smooth noise with bounded log-density derivative and a bounded derivative of each pure order, posterior derivatives can be bounded directly by likelihood-ratio differentiation; bounded h_C then supplies domination by a constant times f . For Laplace noise, the same calculation holds away from the finite shifted kinks, a null set in z , and the bounded log slope again supplies domination.

For $\bar{e}_\theta(p) = \int f(z)H_C(B_{p,r}(\mu_\theta(z;p))) dz$, dominated differentiation gives

$$\frac{d\bar{e}_\theta}{dp} = \int f(z)h_C(B_{p,r}(\mu_\theta)) \left[-\frac{p}{r} + (g_H - g_L)\frac{\partial\mu_\theta}{\partial p} \right] dz. \quad (\text{OA.54})$$

With $E = (\bar{e}_H + \bar{e}_L)/2$ and $\mathcal{R}_T = t_0 + \sum_\theta \bar{e}_\theta(t_\theta - t_0)/2$, the product rule gives

$$\frac{d\mathcal{R}_T}{dp} = (1 - E) \left(1 - \frac{2p}{r} \right) + E \frac{p}{r} + \frac{1}{2} \sum_\theta \frac{d\bar{e}_\theta}{dp} (t_\theta - t_0). \quad (\text{OA.55})$$

The calculation uses $\partial_p t_0 = 1 - 2p/r$, $\partial_p t_H = \partial_p t_L = p/r$, and $\partial_p B|_\mu = -p/r$. At an atomic preparation threshold or a nonregular continuation, this differentiation argument is unavailable. The level objective and full conditional laws remain the correct objects.

A seller equilibrium requires a feasible continuation selection after each reserve and a maximizing reserve under that selection. The numerical envelope of found continuations is an exploration of this problem, not a proof of existence, measurable selection, global attainment, or optimality. These are the explicit open parts of the sale-design result.

B. Computer-assisted equilibrium certificates

This appendix explains what the computer proves about the asymmetric equilibria of Proposition 3, and how. The point deserves to be stated plainly. A floating-point root of the equilibrium equation is not a proof. What I certify is an exact root inside a stated bracket and a global best response for the other trader type, with every rounding error accounted for by interval arithmetic.

B.1. Objects to be certified

The object certified is the existence of an equilibrium, not an approximate fixed point. Fix the benchmark primitives and a strength r . Candidate orders are $(q_H, q_L) = (1, -v)$ with $v \in (0, 1)$. Let $\tau = (c_H - g_L)/(g_H - g_L)$, $c = (1 - v)/2$, and

$$\mu_v(x) = \text{logistic} \left(\frac{|x + v| - |x - 1|}{b} \right), \quad x^*(r, v) = \frac{b \logit \tau + 1 - v}{2}. \quad (\text{OA.56})$$

The certificate region has $1/2 < \tau < M_v$, where $M_v = \text{logistic}((1 + v)/b)$. Thus x^* is strictly between $-v$ and 1. Set $e(x) = \rho + (1 - \rho)\mathbf{1}\{x \geq x^*\}$ and define $A_H = e\Delta_T(1 - \mu_v)$, $A_L = e\Delta_T\mu_v$.

For a signed direction $\epsilon \in \{-1, 1\}$ and magnitude $s \in [0, 1]$, the investor's per-unit convolution and marginal profit are

$$F_\epsilon(s) = \int f(x - \epsilon s) A_\epsilon(x) dx, \quad U'_\epsilon(s) = F_\epsilon(s) + sF'_\epsilon(s) - k, \quad (\text{OA.57})$$

where $A_+ = A_H$ and $A_- = A_L$. In the root condition $\Psi(r, v) = U'_-(v; 1, -v)$, the derivative is with respect to the deviating magnitude s , holding the candidate schedule fixed. The outer change of v recomputes that schedule. Confusing these derivatives would solve the wrong equilibrium condition.

B.2. Global low-type optimality using a Stieltjes measure

The Laplace location family under $1 > -v$ gives a nondecreasing posterior, strictly increasing on $(-v, 1)$. Because $e(\mu)$ is nondecreasing and bounded below by ρ , the function A_L is bounded, nondecreasing, and nonconstant. Its right-continuous representative defines a finite nonzero positive measure dA_L by $dA_L((x, y]) = A_L(y) - A_L(x)$.

Integration by parts on finite intervals, followed by passage to the infinite endpoints, gives

$$F'_L(s) = \int f'(x+s)A_L(x) dx = - \int f(x+s) dA_L(x). \quad (\text{OA.58})$$

The boundary term vanishes because A_L is bounded and the Laplace density tends to zero in both tails. The measure integral is finite because f is bounded and dA_L has finite mass. It is strictly positive before the minus sign because $f > 0$ and the measure is nonzero. An entry jump contributes its positive atom to this measure.

The function f is globally Lipschitz. Therefore $s \mapsto \int f(x+s)dA_L(x)$ is Lipschitz, implying that F'_L is absolutely continuous and differentiable almost everywhere. Except when $-s$ hits an atom of dA_L , its derivative follows by dominated differentiation. The exceptional set of such atoms is at most countable, so

$$F''_L(s) = - \int f'(x+s) dA_L(x), \quad |F''_L(s)| \leq \frac{1}{b} \int f(x+s) dA_L(x) = -\frac{1}{b} F'_L(s) \quad (\text{OA.59})$$

almost everywhere. Hence

$$U''_L(s) = 2F'_L(s) + sF''_L(s) \leq (2 - s/b)F'_L(s) < 0 \quad (\text{OA.60})$$

for $b > 1/2$ on the unit order interval. Since U'_L is absolutely continuous, this strict almost-everywhere inequality makes it strictly decreasing. Any interior root is therefore the unique global maximizer of U_L over all correctly signed magnitudes, including the endpoints.

B.3. Continuity of the equilibrium root equation

Let $[v_-, v_+]$ lie strictly inside $(0, 1)$ and preserve $1/2 < \tau < M_v$. The threshold $x^*(r, v)$ is continuous in v . For any convergent sequence of candidate magnitudes, the posterior and the entry residuals converge pointwise except at the limiting entry boundary. The density and its derivative are translated by bounded amounts; they are dominated by constant multiples of $f(x)$, with an additional factor $1/b$ for the derivative. The residuals are bounded by Δ_T . Dominated convergence therefore gives continuity of both $F_L(v; v)$ and its unilateral derivative component, except for individual density kink points that have zero Lebesgue mass and do not change the integral. Thus $\Psi(r, v)$ is continuous.

Outward enclosures satisfying

$$\inf \Psi(r, v_-) > 0, \quad \sup \Psi(r, v_+) < 0 \quad (\text{OA.61})$$

prove existence of an exact root $v^* \in (v_-, v_+)$. The concavity result proves optimality at that root against its own price schedule. It does not prove that Ψ has only one root across all possible candidate schedules; no

such uniqueness is needed for Proposition 3.

B.4. A global cover for favorable-information deviations

The distributional second derivative of the Laplace density is

$$f'' = \frac{f}{b^2} - \frac{\delta_0}{b^2}. \quad (\text{OA.62})$$

Away from zero, ordinary differentiation gives $f'' = f/b^2$. The first derivative jumps from $1/(2b^2)$ to $-1/(2b^2)$ at zero, contributing the atom $-\delta_0/b^2$. This verifies (OA.62), for example by integrating twice against a compactly supported smooth test function.

Convolving with a bounded measurable A_H yields $F_H'' = (F_H - A_H)/b^2$ as distributions and almost everywhere as functions. Since $F_H, A_H \in [0, \Delta_T]$, the weak second derivative is bounded by Δ_T/b^2 in absolute value. The first derivative has an absolutely continuous, Lipschitz representative; combined with $|F_H'| \leq \Delta_T/b$, this gives

$$|U_H''(s)| \leq L_U = \frac{2\Delta_T}{b} + \frac{\Delta_T}{b^2}, \quad s \in [0, 1] \text{ a.e.} \quad (\text{OA.63})$$

For $s_j = j/n$, evaluate interval enclosures of $U_H'(s_j)$ **uniformly for every v in the root bracket**. Every point of the order interval lies within $1/(2n)$ of a mesh point, so the global bound is

$$\Gamma_H := \min_j \inf U_H'(s_j; [v_-, v_+]) - \frac{L_U}{2n}. \quad (\text{OA.64})$$

A strictly positive lower enclosure for Γ_H proves that buying the maximum amount is uniquely optimal. Checking only the numerical root midpoint does not suffice: the exact root is known only to lie inside the bracket. The interval cover must hold for the whole bracket. All wrong-signed orders are dominated by zero by the residual signs already proved.

B.5. Elementary antiderivatives and infinite tails

Split the integration domain at $-v$, x^* , 1 , and the density center $\zeta = \epsilon s$. Within the posterior's central region let $t = e^{(x-c)/b}$, so $\mu_v = t^2/(1+t^2)$ and $dx = b dt/t$. I use the following primitives, each verified by differentiation:

$$\begin{array}{c|cc} & \int e^{x/b}(\cdot) dx & \int e^{-x/b}(\cdot) dx \\ 1 - \mu_v & be^{c/b} \arctan t & be^{-c/b}(-t^{-1} - \arctan t) \\ \mu_v & be^{c/b}(t - \arctan t) & be^{-c/b} \arctan t \end{array} \quad (\text{OA.65})$$

For $x < \zeta$, the density multiplier is $e^{-\zeta/b}e^{x/b}/(2b)$; for $x > \zeta$ it is $e^{\zeta/b}e^{-x/b}/(2b)$. Multiply the relevant primitive by this constant, by $\epsilon\Delta_T$, and evaluate it at the segment endpoints. On each segment the marginal-density multiplier for F'_ϵ is $\epsilon \operatorname{sgn}(x - \zeta)/b$, so the same segment integral gives the derivative. This avoids differentiating the candidate entry threshold during a unilateral deviation.

Outside $[-v, 1]$ the posterior is constant. For a segment $[a_0, a_1]$ left of ζ , its density mass is $[e^{(a_1-\zeta)/b} - e^{(a_0-\zeta)/b}]/2$; right of ζ it is $[e^{-(a_0-\zeta)/b} - e^{-(a_1-\zeta)/b}]/2$. The infinite endpoint terms vanish analytically.

Multiplying by the segment's constant residual integrates both infinite tails without truncation.

Endpoint ordering must be certified. When the low trader is evaluated at $s = v$, the center is identically $-v$ and must be coalesced algebraically, not treated as two uncertain endpoints. The same applies at $s = 1$ for the high trader. For distinct uncertain endpoints, prove the ordering by disjoint interval bounds. If an interval center overlaps an entry boundary or another cut, subdivide the parameter box or the order cell until the ordering can be proved; do not select an order using only midpoints.

B.6. Certificate acceptance and what it establishes

I use exact decimal input strings and interval operations rounded outward. In an interval test, the infimum and supremum of a displayed function evaluation mean the endpoints of its outward interval enclosure; the underlying equilibrium function remains scalar-valued. The starting precision and mesh are specified in Appendix C.2. An enclosure must retain its full endpoint representation; a printed floating-point midpoint is not the certificate. For a bracket to establish the stated result, I require all of the following: the model support inequalities, low-cost participation and high-cost prior exclusion, strict threshold ordering throughout the bracket, opposite enclosed endpoint signs in (OA.61), positive Γ_H in (OA.64), and a positive pooling-existence margin at the same strength. Entry is enclosed by evaluating the exact tail formula throughout the bracket. Pairwise strictly ordered entry intervals establish the cross-economy increase.

Failure of an enclosure to exclude zero is an unresolved certificate, not evidence of a profitable deviation or nonexistence. A failed sign in a claimed successful certificate is an acceptance failure. Raising precision or refining a bracket is legitimate only with the complete failed and successful record retained.

The result is computer-assisted existence for exact equilibria inside specified intervals. It does not prove uniqueness of the informative profile, exclude mixed equilibria, or certify an entire continuous branch between nodes. A continuation curve between certified nodes is a numerical diagnostic until additional interval and continuation arguments establish it. This distinction governs the correspondence figure.

C. Numerical exercises and quantity registry

C.0. Input declarations, output conventions, and acceptance

This section is the complete numerical contract. The literal values in its input declarations specify experiments; the manuscript's quantitative placeholder fields are filled only by validated outputs or these declared inputs. Mathematical constants, equation and result labels, citation metadata, and software identifiers are structural, not estimated quantities. The empirical scaffold contributes no sample size or empirical estimate to the registry.

All parameters below are exact decimal inputs, converted without a binary-floating intermediate when interval certification is required. A numerical result records its full parameter vector, noise law, cost law, information structure, sale institution, tie convention, and branch. Defaults must not silently cross from one exercise to another. Scalar probabilities are stored as probabilities; percentage formatting is a separate presentation step.

Input declaration: benchmark.

$h = 10$; $e_{ll} = 1$; $p = 0.5$; $\rho = 0.25$;

```

c_L = 1; c_H = 6; b = 2; k = 0.02;
r_weak = 1.2; r_strong = 3; r_collapse = 3.6;
noise = Laplace; cost = atomic; fundamental_prior_H = 0.5;
entry_at_indifference = yes; bid_equal_reserve_is_admissible = yes;
cost_halfwidth = 0.1; value_band_halfwidth = 0.05;
binary_alternative_reserve = 1.01; atomless_alternative_reserve = 1.1.

```

Input declaration: moderate acquisition values.

```

h = 2; ell = 1; p = 0.5; rho = 0.25;
c_L = 0.3; c_H = 0.89; b = 2; k = 0.002;
r_weak = 1.05; r_strong = 1.5;
noise = Laplace; cost = atomic; fundamental_prior_H = 0.5.

```

Input declaration: complementary signals.

```

h = 10; ell = 1; p = 0.5; rho = 0.85;
c_L = 1; c_H = 7.14; b = 2; k = 0.015;
r_weak = 1.1; r_strong = 2.3; a = 0.70; d = 0.75;
noise = Laplace; cost = atomic; fundamental_prior_H = 0.5;
T_and_Y_independent_given_fundamental = yes.

```

Input declaration: numerical controls.

```

quadrature_absolute_target = 1e-11;
quadrature_relative_target = 1e-11;
probability_acceptance = 1e-8;
price_identity_acceptance = 1e-8;
entry_optimality_acceptance = 1e-8;
deviation_gain_acceptance = 1e-7;
independent_formula_acceptance = 1e-9;
initial_order_intervals = 400;
refined_order_intervals = 800;
initial_flow_halfwidth = 40;
interval_decimal_precision = 50;
certificate_derivative_intervals = 200;
certificate_refined_intervals = 400;
probability_display_decimals = 6;
parameter_display = shortest_exact_decimal;
certificate_display = outward_interval;

```

The flow halfwidth determines a diagnostic mesh, not an integration truncation. Integrals are evaluated over the full line, or their omitted tails are explicitly bounded. For a bounded residual $A \leq \bar{A}$ and $|q| \leq 1$, a Laplace truncation outside $[-T, T]$, $T > 1$, loses at most $\bar{A}e^{-(T-1)/b}$ in per-unit gross profit. Multiply by the order magnitude for total payoff. Under logistic noise a valid bound is $2\bar{A}/[1 + e^{(T-1)/b}]$. Include these bounds in the error budget; do not compare a tail-truncated integral with a full integral as though both were exact.

I maintain distinct tolerances for equations, quadrature estimates, and strategic deviations. A root or a

finite-grid maximum does not establish equilibrium. For each candidate, evaluate

$$\epsilon_P = \sup_{x \text{ tested}} |P(x) - \mathbb{E}[V_T \mid X = x]|, \quad \epsilon_e = \sup_{(P,C) \text{ tested}} [\text{profit from reversing entry}]_+, \quad (\text{OA.66})$$

and

$$\epsilon_q = \max_{\theta} \left\{ \sup_{q \in [-1,1]} U_{\theta}(q; \sigma) - \int U_{\theta}(q; \sigma) d\sigma_{\theta}(q) \right\}. \quad (\text{OA.67})$$

The tested price supremum is a numerical diagnostic unless a uniform bound is supplied. The order supremum must be bounded globally for a computer-assisted label; a finite approximation is identified separately. Check all wrong-signed orders as well as correctly signed ones in numerical diagnostics, even when an analytical sign argument excludes them. For mixed profiles, every positive-weight support action must attain the same maximal payoff within the stated tolerance. Recompute the price and entry schedules once per candidate, then hold them fixed during each unilateral-deviation calculation.

Mandatory probability identities are $\int a_{\theta} = 1$, $\mathbb{E}\mu_X = 1/2$, and $\mathbb{E}P = \mathbb{E}V_T$. Signal exercises add the joint-posterior identities in A.7. Independent auction integration must match closed forms; independent cost-based and flow-based entry integrations must agree. Repeat numerical diagnostics after doubling the order resolution and tightening both integration targets by a factor of 10, recorded in the run manifest. A claimed accepted row that breaches any acceptance bound terminates validation. Search candidates that genuinely fail are retained as rejected rows; inability to resolve a node is recorded as open, not suppressed or interpolated.

CSV is the boundary between numerical calculations and presentation. Files use UTF-8 and quoted headers when a mathematical symbol contains punctuation. Parameter columns use the symbols `h`, `ell`, `p`, `rho`, `c_L`, `c_H`, `b`, `k`, `r`; additional columns such as `Delta_T`, `B_prior`, `q_H`, `q_L`, `e_H`, `e_L`, `E`, `O_H`, `R_T`, `tau`, `x_star` are transliterations of the paper's notation. The data dictionary below each exercise defines every additional symbol. A branch identifier is a data label, not an economic selection rule.

The quantity registry has columns `name`, `value`, `lower`, `upper`, `units`, `display`, `exercise`, `parameter_set`, `branch`, `status`, `source_file`, `source_row`, `definition`. The name is the exact placeholder key without brackets. Each name is unique. A missing or failed quantity remains unresolved; it is never replaced with zero, a cached value from another parameter set, or a manually typed number. Input declarations can be echoed into the registry with status `input`; derived research quantities use the result-status vocabulary and record their validation. This administrative input type is not an additional result status.

C.1. Baseline, controls, extensions, and scalar reproduction

Inputs. Use the benchmark declaration at weak, strong, and collapse strengths. Evaluate the atomic-cost model with Laplace noise first. Then replace the noise law by logistic noise, retaining the scale parameter, and replace cost atoms by a mixture of uniforms on $[c_L - \varepsilon_C, c_L + \varepsilon_C]$ and $[c_H - \varepsilon_C, c_H + \varepsilon_C]$. Their mixture weights remain $\rho, 1 - \rho$. These replacements define separate economies.

Acquisition layer. Compute the pointwise auction outcomes from (OA.50). Integrate independently over R , splitting at p and ℓ , and verify $t_0, t_H, t_L, g_H, g_L, \Delta_T$. Compute $B_r(m), B_r(1/2), B_r(M)$. Theorem

margins are

$$\begin{aligned}\zeta_L &= B_{r_1}(m) - c_L, & \zeta_{H0} &= c_H - B_{r_0}(1/2), & \zeta_{H1} &= B_{r_1}(M) - c_H, \\ \zeta_0 &= k - \Delta_T(r_0), & \zeta_1 &= (1 - 1/b)\rho m \Delta_T(r_1) - k.\end{aligned}\tag{OA.68}$$

The collapse node additionally requires $c_H - B_{r_2}(M) > 0$, $B_{r_2}(m) - c_L > 0$, and its full-order margin. For atomless costs replace the appropriate cost endpoints as in Proposition A.6. Verify all margins before assigning an analytical equilibrium label.

Financial and entry layer. Pooling has $q_H = q_L = 0$, $\mu = 1/2$, $e = \rho$, and $P = t_0 + \rho[(t_H + t_L)/2 - t_0]$. Full orders use the conditional flow densities and posterior in A.4 or A.6. Compute τ , the appropriate threshold, conditional entry, ownership, mean price, and revenue. Always handle threshold regions using the cost comparison: a cutoff above the feasible posterior means no high-cost entry; equality at a Laplace plateau uses the specified tie rule. Do not insert an infeasible threshold into an interior tail formula.

For atomless costs, entry at a posterior is the complete mixture CDF $H_C(B_r(\mu))$. Independently compute entry by integrating the tail probability for each cost within each uniform component. The two integrals must agree. Compute expected paid preparation costs as well as entry probability; they are not mean cost multiplied by unconditional entry when the high-cost component is selected.

Controls. The frozen-profile control uses full orders at both strengths and reoptimizes preparation and prices, but explicitly does not require that the weak investor chooses that profile. The price-hidden control reoptimizes investor behavior and pricing, while the buyer uses the prior: pooling at weak strength and full orders at strong strength follow their respective bounds. The matched-dividend control adds exactly $\mathcal{R}_T^{\text{feedback}} - \mathcal{R}_T^{\text{hidden}}$ to the hidden financial payoff and price. Verify that all residuals, orders, and entry probabilities are unchanged. It is not an additional acquisition payment.

Welfare. At strong strength, evaluate (OA.47) or (OA.48), and independently subtract total net allocation surplus in (OA.49). Also compute the revenue difference directly from conditional entry and from the difference in mean prices. Store these as different quantities. A strength comparison is not labeled a welfare effect of access to prices.

Validation. For each equilibrium candidate, integrate residual profits over the full noise line, splitting at density centers, posterior kinks, and entry thresholds. Check all orders on the declared initial and refined grids. Include the special points at which a density center crosses an entry boundary. The analytical margins provide the global result for the specified equilibria; the grid is an implementation check. Verify posterior inversion from the observed price, including atom probabilities in flat tails. In the frozen-profile control retain any profitable investor deviations as expected counterfactual diagnostics rather than mistakenly treating that profile as equilibrium.

Outputs.

tables/auction_primitives.csv:

parameter_set, r, p, t_0, t_H, t_L, g_H, g_L, Delta_T, B_m, B_prior, B_M

tables/equilibrium_controls.csv:

parameter_set, noise, cost_law, r, experiment, q_H, q_L, e_H, e_L, E, O_H, R_T, W, expected_preparation_cost, tau, x_star, matched_dividend, status, accepted

tables/extensions.csv:

parameter_set, noise, cost_low, r_weak, r_strong,
E_weak, E_strong, O_H_weak, O_H_strong,
zeta_L, zeta_H0, zeta_H1, zeta_0, zeta_1, status, accepted

numerics/baseline_deviations.csv:

parameter_set, experiment, noise, cost_low, r, state, q,
U(q), U(candidate), deviation_gain, quadrature_error, tail_bound

numerics/feedback_comparisons.csv:

parameter_set, noise, cost_low, r,
W_feedback, W_hidden, W_gain,
R_T_feedback, R_T_hidden, R_T_gain, matched_dividend,
welfare_identity_error, revenue_identity_error,
residual_invariance_error, status, accepted

figures_data/two_returns.csv:

r, mu, Delta_T, B_r(mu), d_Delta_T_dr, d_B_r_mu_dr

W denotes allocation value net of paid preparation costs, not target revenue. The derivatives in the last file are the explicit uniform formulas. Use the correspondence strength grid from C.2 and beliefs $m, 1/2, M$. These outputs feed Tables 1–3 and Figure 2. Scalar keys are defined in C.8; no displayed value is copied directly from a figure.

C.2. The equilibrium correspondence and interval certificates

Inputs and grid. Use the benchmark primitives. The initial strength mesh is the exact decimal arithmetic progression from 1.005 through 3.800 by 0.005. Add the weak, strong, collapse, and certified strengths, and the exact boundaries $r(k), r_N, r_U, r_C$. Evaluate both sides of each threshold using offsets 0.0001 and 0.001, while treating the equality at r_C symbolically as $\tau = M$ rather than replacing it by a rounded numerical root. Refine intervals with branch changes or solver disagreements to strength spacing 0.001 or smaller and record the refinement rule.

The certificate input brackets are exact decimals:

$r = 1.55$; $v_{\text{left}} = 0.46031618$; $v_{\text{right}} = 0.46031620$;
 $r = 1.60$; $v_{\text{left}} = 0.70747537$; $v_{\text{right}} = 0.70747539$;
 $r = 1.65$; $v_{\text{left}} = 0.90333198$; $v_{\text{right}} = 0.90333201$.

Analytically classified outcomes. First verify membership in the maintained floor/prior domain at each node. Pooling existence is given by $k - \rho\Delta_T/2 \geq 0$ there; its sufficient uniqueness bound is $k - \Delta_T > 0$. Full-order uniqueness is established by $(1 - 1/b)\rho m\Delta_T - k > 0$. Record these margins individually. Compute r_N, r_U, r_C from A.5 and verify each defining equation, including support restrictions. Outside a maintained domain, evaluate the actual prior entry rule and continuation instead of retaining a classification based on ρ .

Full-order candidates. At every remaining node, form the full-order posterior and entry policy and

globally check each trader's best response. The sharper candidate-specific J test in (OA.22) can establish existence. Failing that sufficient test does not reject the profile; use an interval payoff or derivative search. For the low type, strict concavity reduces the full-order check to $U'_L(1) \geq 0$. For the high type, a positive derivative certificate is sufficient but not necessary; if it fails, bound the global payoff gap directly, splitting around all stationary points and kinks. Retain a numerical diagnostic unless every between-grid gain is enclosed.

Asymmetric continuation. At each r , solve $\Psi(r, v) = 0$ for candidates with $(1, -v)$ and $v \in (0, 1)$. Bracket every observed sign change on an initial magnitude mesh of spacing 0.01 , add the certified brackets, and adapt near small residuals and threshold boundaries. A root that does not change sign can exist, so use additional local minimization of $|\Psi|$ and retain unresolved tangencies rather than claiming that sign-change enumeration is exhaustive. Continue candidate roots in both increasing and decreasing strength, using the previous solution only as one initialization. A fixed-point continuation must distinguish the unilateral derivative in s from the change in the candidate schedule as v changes.

At each candidate compute x^*, e_H, e_L, E, O_H , revenue, the pooling-existence flag, and all global best-response diagnostics. At selected nodes apply Appendix B: enclose the root, prove its endpoint signs, verify low-type concavity, and cover high-type deviations uniformly in the root bracket. Reproduce the declared certificates first. Additional certified nodes are added where interval ordering and a positive high-type margin can be established. A root approaching $v = 1$ is a branch boundary, not permission to extend an interior equation beyond its domain.

Other pure and mixed candidates. The complete correspondence is an open characterization. The numerical search must nevertheless avoid restricting every informative outcome to the asymmetric family above. Search pure profiles $(u, -v)$ over $[0, 1]^2$, including both interior magnitudes, using multiple initializations and global unilateral maximization. Search mixed profiles on adaptive order supports: start with correctly signed meshes of spacing 0.05 , allow independent probabilities for each type, solve support-payoff indifference and probability-simplex constraints, then add any profitable off-support actions. Repeat after reducing the mesh spacing. Zero-probability support actions are removed only with their residuals retained. No finite support search proves that other supports or mixed equilibria do not exist.

A mixed candidate with support points $q_{\theta j}$ and weights $\omega_{\theta j}$ has conditional density $a_{\theta}(x) = \sum_j \omega_{\theta j} f(x - q_{\theta j})$. Recompute its posterior, entry, and competitive price. Its payoff is the weighted average of the support payoffs. All positive-weight points must be best responses within tolerance, and every off-support deviation must be checked. Preserve distinct roots and supports found from different initializations. A candidate search that fails to resolve a node receives an open flag; it does not create an empty equilibrium set.

Figure discipline. Plot analytical uniqueness regions separately from existence-only regions. Show every accepted observed branch. Certified points have interval bars. Numerical lines are broken at failed validation, discontinuities, branch changes, or unresolved gaps. The multiplicity flag means that distinct equilibria have been established or found at a node, not that all equilibria have been enumerated. The ordered certified entry intervals provide the stated cross-economy comparison; a fitted derivative on an exploratory line is not substituted for that result.

Outputs.

numerics/correspondence.csv:

r , branch, q_H , q_L , v , e_H , e_L , E , O_H , R_T , τ , x_{star} ,

pooling_exists, pooling_unique_bound, full_unique_bound,
existence_status, uniqueness_status, accepted, multiplicity_found,
epsilon_P, epsilon_e, epsilon_q, tail_bound, unresolved_reason

numerics/mixed_supports.csv:

r, branch, state, support_index, q, weight, U(q), support_gap,
off_support_gain_bound, accepted, status

numerics/certificates.csv:

r, v_lower, v_upper, Psi_left_lower, Psi_left_upper,
Psi_right_lower, Psi_right_upper, Gamma_H_lower,
L_U_upper, mesh_intervals, interval_digits,
E_lower, E_upper, pooling_margin_lower, threshold_margin_lower, accepted

numerics/thresholds.csv:

boundary, value, lower, upper, defining_residual,
in_support_domain, low_cost_floor_valid, interpretation

The threshold file uses exact row labels `pooling_unique_sufficient`, `pooling_existence`, `full_orders_unique_sufficient`, and `high_cost_ceiling`. It also carries separately labeled analytical scalars `m`, `M`, and `laplace_entry_left_limit` needed by the registry; these are not mislabeled activation thresholds. Every row stores its defining expression and checks. Fields that do not apply to a probability scalar are marked not applicable, not set to zero.

All general parameter columns are included or keyed to an immutable parameter declaration whose full contents accompany the file. `uniqueness_status` distinguishes analytical uniqueness from not established; it is never inferred from the number of search hits. Feed: Figure 1, Proposition 3, the threshold discussion, and the baseline table. Registry keys `cert_a_*`, `cert_b_*`, and `cert_c_*` refer to the ordered declared nodes, not arbitrary roots selected from a larger search.

C.3. Complementary private signals

Inputs. Start with the complementary-signal declaration. Then cross trader accuracies $\{0.68, 0.69, 0.70, 0.71, 0.72\}$ with buyer accuracies $\{0.73, 0.74, 0.75, 0.76, 0.77\}$, holding other primitives fixed. These are separate equilibria with exact decimal parameters, not a claim that every grid point meets Proposition A.7.

Objects. Compute m , M , public fundamental bounds μ_- , μ_+ , the worst joint buyer posterior $\phi_-(\mu_-)$, and the best joint posterior $\phi_+(\mu_+)$. Compute the five strict margins from (OA.29): the low-cost floor, high-cost exclusion using the buyer's favorable private signal at weak strength, high-cost profitability after a favorable strong public price and private signal, weak residual exclusion, and strong global derivative margin. If a margin fails, withhold the analytical theorem label; do not conclude that the economic reversal is absent.

At both strengths construct signal-contingent order laws and the correct state-dependent entry policy. For the theorem-supported weak node public belief is the prior and both high-cost private-signal types stay out. For the strong node use full orders by T , then compute the public posterior from λ_X , each buyer posterior

$\phi_Y(\mu_X)$, and both private-signal cutoffs in (OA.39). Always handle both $Y = +$ and $Y = -$, including always/never and plateau-equality cases. The benchmark example's exclusion of one private-signal type is an outcome, not an imposed simplification for the sweep.

Independent verification. First integrate entry using (OA.40)–(OA.41). Independently sum over the finite fundamental, trader-signal, and buyer-signal states and integrate Z and C . Compute the price directly as the conditional expectation of V_T given flow and compare it with (OA.35). Recover the public posterior from the price inversion and then update using Y ; compare with the direct joint likelihood. Verify the residual identities (OA.37) by direct conditional trader payoffs. Finally evaluate all unilateral orders using the fixed candidate schedules. The investor must not be given the buyer's private signal during this calculation.

Outputs.

numerics/two_signals.csv:

a, d, r, q_plus, q_minus, mu_lower, mu_upper,
 phi_minus_mu_lower, phi_plus_mu_upper,
 x_star_Yplus, x_star_Yminus, e_H, e_L, E, O_H, R_T,
 low_cost_margin, private_only_exclusion_margin, joint_entry_margin,
 weak_order_margin, strong_order_margin,
 posterior_error, residual_error, epsilon_P, epsilon_q,
 status, accepted

numerics/two_signal_deviations.csv:

a, d, r, trader_signal, q, U(q), U(candidate),
 deviation_gain, quadrature_error, tail_bound

The same complete primitive vector accompanies each row. The scalar-margin names map to these columns in order: low cost to `low_cost_margin`, high prior to `private_only_exclusion_margin`, high ceiling to `joint_entry_margin`, weak trade to `weak_order_margin`, and strong trade to `strong_order_margin`. These are comparison-level margins; repeat them consistently on both strength rows or store them in a separately keyed comparison record. The display accuracy keys are percentages generated from the raw probability keys, never separately typed. Table 3 uses the declared example and marks the sweep's condition region separately. Feed: the example of Proposition A.7 and Table 3.

C.4. Moderate values and constructive nonemptiness

Inputs. Reproduce the moderate declaration exactly. Compute the same primitives, theorem margins, complete equilibrium objects, and controls as in C.1. Use independent auction integration and global analytical margins, not only a best-response mesh, to validate the theorem-supported example.

For the nonemptiness construction, set `e11 = 1`, `p = 0.5`, `rho = 0.25`, `b = 2`, and use high-to-low ratios $\{1.05, 1.10, 1.25, 1.50, 2.00\}$. For each ratio let $D = h - \ell$ and examine $\delta_j = D2^{-j}$ for integer j from 1 through 20. Define $r_1 = \ell + \delta_j$ and $r_0 = \ell + \delta_j^2/D$. Test the conditions needed in A.6. When the intervals are nonempty choose

$$\begin{aligned}
k &= \frac{\Delta_T(r_0) + (1 - 1/b)\rho m \Delta_T(r_1)}{2}, \\
c_H &= \frac{B_{r_0}(1/2) + B_{r_1}(M)}{2}, \quad c_L = \frac{B_{r_1}(m)}{2}.
\end{aligned}
\tag{OA.69}$$

Every chosen value is an output of the construction, not a fixed calibration imposed on all ratios. Check $c_L < c_H$, support, and all theorem inequalities explicitly. Keep the sequence of failed and successful j values. Small margins require adequate arithmetic precision; floating-point cancellation is not evidence against nonemptiness. Report the first certified feasible member of the declared sequence and the numerical scale of k , not an invented uniform lower bound on its size.

Outputs.

numerics/moderate_values.csv:

h, ell, p, rho, c_L, c_H, b, k, r_weak, r_strong,
zeta_L, zeta_H0, zeta_H1, zeta_0, zeta_1,
E_weak, E_strong, O_H_weak, O_H_strong, status, accepted

numerics/nonemptiness_construction.csv:

h_over_ell, j, delta, r_weak, r_strong, k, c_L, c_H,
all_margins_min, feasible, arithmetic_precision, reason

Feed: Table 3 and the moderate-value paragraph. The general construction is an analytical argument; the finite sequence is a diagnostic illustration of its scale.

C.5. Noise laws, threshold distance, and posterior upper tails

Inputs. Fix benchmark strong strength, b , and ρ . Use full orders. Form 401 equally spaced points for the normalized threshold distance $d_\tau = (M - \tau)/(M - 1/2)$ on $[0, 1]$, and add the benchmark τ , $M - 10^{-4}$, $M - 10^{-6}$, and $M + 10^{-6}$. At the endpoints use the stated mathematical limits and tie convention rather than unstable logarithms. Compare Laplace and logistic laws at the same scale parameter; do not claim equal variance or a Blackwell ordering between the laws.

Objects. For each threshold, compute the order-flow cutoff, $\alpha_H(\tau)$, $\alpha_L(\tau)$, posterior upper-tail mass $[\alpha_H + \alpha_L]/2$, and the induced entry $\rho + (1 - \rho)[\alpha_H + \alpha_L]/2$. Verify each tail by direct integration of the conditional flow densities. For the baseline logistic example, compute $x^*/(b\pi/\sqrt{3})$ from the derived noise variance, not from the aggregate-flow variance. Store both variances separately if both standardized cutoffs are reported.

A threshold scan is first an information-experiment comparison. To interpret a row as an equilibrium with changed preparation cost, set $c_H(\tau) = g_L + \tau(g_H - g_L)$, keep the other primitives fixed, and recheck the low-cost floor, cost ordering, and global full-order bound. Label only validated rows as such. At the strong benchmark the full-order bound does not depend on the high-cost threshold, but that fact must not be presumed after changing other primitives.

At $\tau = M$, Laplace admits entry in its entire upper posterior plateau under the tie convention; logistic has zero mass at the unattained endpoint. For $\tau > M$, both tail masses are zero. This distinction is part of the plotted result. On the lower endpoint $\tau = 1/2$, the threshold is zero and no artificial singularity arises.

Outputs.

figures_data/posterior_tails.csv:

```
noise, b, tau, M_minus_tau, normalized_distance, x_star,
alpha_H, alpha_L, posterior_upper_tail_mass, E,
implied_c_H, noise_variance, flow_variance, threshold_noise_sd,
full_order_margin, equilibrium_interpretation_valid,
integration_error, status
```

Feed: Figure 3, the logistic magnitude paragraph, and Table 3's noise rows. Infinite thresholds are stored with an explicit unattainable label, not converted to an arbitrary finite cutoff.

C.6. Reserve comparisons and exploratory seller continuation

Inputs. Reproduce the binary-value comparison using the benchmark reserves 0.5 and 1.01 at strengths 1, 2 and 3. Reproduce the atomless-value class economy using half-width 0.05 and reserves 0.5 and 1.1 at the same strengths. In the class economy the investor sees the class only and preparation reveals the exact value. Do not let the investor trade on the within-class realization.

Payoff oracle. Integrate the realized outcomes (OA.50) over the independent uniform incumbent and each conditional value distribution. Split at $R = p$, $R = v$, and all reserve/value support boundaries. Verify against (OA.51) and, where applicable, (OA.52). Compute class-conditional t_H, t_L, g_H, g_L for every reserve, including reserves inside a value band. A formula for $p < \ell$ or for exclusion of an entire class is not used inside a partially excluded band.

Continuation with and without an entry floor. For a candidate class-contingent order profile, compute its raw flow posterior μ_X . When $\Delta_T > 0$ and there is positive entry, the benchmark inversion applies to the positive-entry prices. If entry vanishes on a set of flows, those flows can pool at $P = t_0$. Compute their joint posterior by integrating the full preimage of that price:

$$\mu(P = t_0) = \frac{\int_{\{x:P(x)=t_0\}} a_H(x) dx}{\int_{\{x:P(x)=t_0\}} [a_H(x) + a_L(x)] dx}. \quad (\text{OA.70})$$

The formula is used only when the denominator is positive. The actual buyer entry rule is based on this price posterior, not on the raw posterior inside that pool. Recheck its consistency and competitive pricing pointwise in flow. A convenient candidate construction for independent costs is $P(\mu) = t_0 + H_C(B(\mu))[t_L - t_0 + \Delta_T \mu]$: it is strictly increasing wherever entry is positive and $\Delta_T > 0$, while zero-entry flows share t_0 . The zero-entry pool remains consistent only after its pooled posterior and the cost comparison are verified. This check is required even when the raw posterior formula looks well behaved.

At reserves with $\Delta_T = 0$, all entry impossible, or a degenerate payoff, analyze the constant-price candidate directly. Pooling uses the actual prior entry probability $e_0 = H_C(B(1/2))$, and its correctly signed deviation coefficient is $e_0 \Delta_T / 2 - k$; it is not automatically $\rho \Delta_T / 2 - k$. When $e_0 = 0$, no trade and no entry are supported by constant pricing. All accepted candidates must satisfy the original conditional pricing and entry conditions.

Specified comparisons. At the declared reserves, verify the relevant strict no-trade or full-order bounds using the class payoff spread and the actual floor. Compute entry and seller revenue and compare each alternative reserve with the original one at the same strength. This reproduces the profitable alternatives, not a global reserve optimum.

Exploratory sweep. For binary values scan reserves from 0 to h in steps of 0.05; for the class economy

scan to $h + \text{epsilon_V}$ at the same spacing. Add exactly $\ell \pm \varepsilon_V$, $h \pm \varepsilon_V$, r , the declared reserves, and the boundary offsets 0.0001 on both sides whenever feasible. Refine intervals with changing trading behavior, unresolved candidates, or candidate revenue maxima to spacing 0.002 or smaller. At each reserve re-solve pooling, full-order, asymmetric, other pure, and finite-support mixed candidates using C.2's best-response discipline and the correct price pools. Warm starts in both directions supplement, rather than replace, independent starts.

For each reserve retain every accepted continuation found. Report the minimum and maximum entry and revenue across those continuations and the unresolved-search flag. These are found-continuation ranges, not certified envelopes of all equilibria. Never replace a failed solve with the nearest successful reserve, optimize a frozen information experiment, or label the best sampled reserve globally optimal. Any proposed seller equilibrium must specify its continuation at unchosen reserves as well as chosen ones.

Outputs.

tables/reserve_comparisons.csv:

value_low, signal_information, epsilon_V, r, p,
t_0, t_H, t_L, g_H, g_L, Delta_T,
q_H, q_L, e_H, e_L, E, R_T,
low_cost_floor_margin, trading_margin, payoff_oracle_error,
status, accepted

numerics/reserve_continuations.csv:

value_low, r, p, branch, q_H, q_L,
E, O_H, R_T, no_entry_price_mass, posterior_in_no_entry_pool,
epsilon_P, epsilon_e, epsilon_q, status, accepted, unresolved_reason

numerics/reserve_ranges.csv:

value_low, r, p, accepted_continuations_found,
E_min_found, E_max_found, R_T_min_found, R_T_max_found,
search_unresolved, global_envelope_certified

The final column is false unless an additional exhaustive argument establishes the full envelope. Feed: Table 4 and the sale-design discussion. The research question about seller-optimal discovery is not assigned an answer by this exploratory file.

C.7. Bargaining weights and the payment property

Inputs. Use the benchmark h, ℓ and weak/strong incumbent distributions, but set the bargaining institution's reserve to zero as specified in Proposition A.8. Evaluate seller weights from 0 to 0.99 by 0.01 , including the exact midpoint 0.5 . The endpoint $\eta = 1$ can be stored separately as a payment-stage limit, not as a positive-cost entry equilibrium.

Objects and verification. Compute $T_\eta(R, v)$ and the winning challenger's profit pointwise, then integrate over the incumbent. Independently compare the results with (OA.44), including no-entry revenue $\eta r/2$. Verify feasibility of each transfer between fallback and winning value and the Nash first-order condition for interior weights. Compute differences between strong and weak distributions for Δ_η and $G_{\theta, \eta}$. The spread

difference must change sign at the midpoint, with zero at that exact weight. This is an acquisition-stage exercise; do not append a trader or entry prediction without solving the new continuation game.

Outputs.

figures_data/bargaining.csv:

eta, r, t_0_eta, t_H_eta, t_L_eta, Delta_eta,
G_H_eta, G_L_eta, spread_strength_difference,
profit_H_strength_difference, profit_L_strength_difference,
transfer_feasibility_error, integration_error, status

Feed: Figure 4 and Proposition A.8. Parameter values and the change of institution are stated in the figure caption.

C.8. Complete scalar registry and placeholder substitution

Every quantitative placeholder is defined below. I distinguish exact input declarations from computed quantities and interval enclosures. The declaration is the numerical input; the unresolved placeholder in the manuscript is not another input. A replicator reads the declared values in C.0 and the tables below, computes the specified row, validates it, and only then substitutes its formatted value. Mathematical constants, equation and result numbers, bibliographic identifiers, and data-schema labels remain literal: they are not outputs of a numerical exercise.

The output registry is `numerics/quantity_registry.csv`, with the schema in C.0. The accompanying `quantity_manifest.csv` repeats the definitions below; it is a specification, not a file of prefilled results. Each derived row records its source file and an unambiguous row selector. The source row must itself identify the full parameter vector, information regime, cost and noise law, and continuation branch. A scalar referring to a difference records both rows used to form the difference. An unresolved source leaves its placeholder unfilled and records the reason; the renderer must reject a request for a fully filled manuscript in that state.

Display conventions are fixed as follows. `exact_input` prints the input decimal without changing its value. `decimal_6` and `decimal_9` round ordinary scalar diagnostics to six and nine decimal places. `scientific_10` prints ten significant digits in scientific notation. `percent_integer` displays the underlying probability as an integer percentage. `outward_interval_8` and `outward_interval_10` round lower endpoints downward and upper endpoints upward to the indicated decimal precision. `lower_bound_10` rounds a lower bound downward to ten decimal places. Interval outputs in display mathematics are inserted as LaTeX brackets; percentage outputs include the percent sign. Rounding is performed only after all validations on unrounded values. A displayed lower bound must remain strictly positive to support an asserted strict inequality.

The substantive statuses attached to output rows inherit the result actually supported. An evaluation satisfying the analytical theorem region records that region and its margins. A computer-assisted row requires every certificate predicate, not a successful floating-point root. A finite search outside proved regions remains a numerical diagnostic. A failed or unresolved problem remains open. The declaration records carry the administrative status `input`.

Declared scalars. These values identify the input vector or execution metadata, rather than replace any derived result.

Placeholder key	Exact declaration	Meaning
base_h	10	Declared h ; retain its exact decimal input.
base_ell	1	Declared ℓ ; retain its exact decimal input.
base_p	0.5	Declared p ; retain its exact decimal input.
base_rho	0.25	Declared ρ ; retain its exact decimal input.
base_c_low	1	Declared c_L ; retain its exact decimal input.
base_c_high	6	Declared c_H ; retain its exact decimal input.
base_b	2	Declared b ; retain its exact decimal input.
base_k	0.02	Declared k ; retain its exact decimal input.
base_r_weak	1.2	Declared r_0 ; retain its exact decimal input.
base_r_strong	3	Declared r_1 ; retain its exact decimal input.
base_r_collapse	3.6	Declared r_2 ; retain its exact decimal input.
moderate_h	2	Declared h ; retain its exact decimal input.
moderate_ell	1	Declared ℓ ; retain its exact decimal input.
moderate_p	0.5	Declared p ; retain its exact decimal input.
moderate_rho	0.25	Declared ρ ; retain its exact decimal input.
moderate_c_low	0.3	Declared c_L ; retain its exact decimal input.
moderate_c_high	0.89	Declared c_H ; retain its exact decimal input.
moderate_b	2	Declared b ; retain its exact decimal input.
moderate_k	0.002	Declared k ; retain its exact decimal input.
moderate_r_weak	1.05	Declared r_0 ; retain its exact decimal input.
moderate_r_strong	1.5	Declared r_1 ; retain its exact decimal input.
signal_h	10	Declared h ; retain its exact decimal input.
signal_ell	1	Declared ℓ ; retain its exact decimal input.
signal_p	0.5	Declared p ; retain its exact decimal input.
signal_rho	0.85	Declared ρ ; retain its exact decimal input.
signal_c_low	1	Declared c_L ; retain its exact decimal input.
signal_c_high	7.14	Declared c_H ; retain its exact decimal input.
signal_b	2	Declared b ; retain its exact decimal input.
signal_k	0.015	Declared k ; retain its exact decimal input.
signal_r_weak	1.1	Declared r_0 ; retain its exact decimal input.
signal_r_strong	2.3	Declared r_1 ; retain its exact decimal input.
signal_trader_accuracy_value	0.70	Declared a ; retain its exact decimal input.
signal_buyer_accuracy_value	0.75	Declared d ; retain its exact decimal input.
cost_halfwidth	0.1	ε_C , the preparation-cost half-width.
value_band_halfwidth	0.05	ε_V , the within-class value half-width.
value_reserve_high	1.1	The class-economy reserve above the entire low-value band.
cert_a_r	1.55	Declared incumbent strength of the corresponding certified node.
cert_b_r	1.60	Declared incumbent strength of the corresponding certified node.
cert_c_r	1.65	Declared incumbent strength of the corresponding certified node.

Placeholder key	Exact declaration	Meaning
seed_python_version	3.13.5	Documented software version for the distributed verification results; a new run records its actual version separately.
seed_numpy_version	2.3.5	Documented software version for the distributed verification results; a new run records its actual version separately.
seed_scipy_version	1.17.0	Documented software version for the distributed verification results; a new run records its actual version separately.
seed_mpmath_version	1.3.0	Documented software version for the distributed verification results; a new run records its actual version separately.

Computed manuscript scalars. All row selectors below refer to the parameter declarations in C.0. Within a parameter set, `r_weak`, `r_strong`, and `r_collapse` select its declared strengths. The basic probability and revenue definitions are $E = (\bar{e}_H + \bar{e}_L)/2$, $O_H = \bar{e}_H/2$, and $\mathcal{R}_T = t_0 + [\bar{e}_H(t_H - t_0) + \bar{e}_L(t_L - t_0)]/2$.

Benchmark and matched controls.

Key	Definition and row selection	Exercise / output	Display
base_entry_weak	$E = (\bar{e}_H + \bar{e}_L)/2$ in the validated Laplace, atomic-cost feedback equilibrium. Select: <code>experiment=feedback; noise=Laplace; cost_law=atoms; r=r_weak</code> .	C.1; tables/equilibrium_controls.csv	decimal_6
base_entry_strong	$E = (\bar{e}_H + \bar{e}_L)/2$ in the validated Laplace, atomic-cost feedback equilibrium. Select: <code>experiment=feedback; noise=Laplace; cost_law=atoms; r=r_strong</code> .	C.1; tables/equilibrium_controls.csv	decimal_6
base_entry_collapse	$E = (\bar{e}_H + \bar{e}_L)/2$ in the validated Laplace, atomic-cost feedback equilibrium. Select: <code>experiment=feedback; noise=Laplace; cost_law=atoms; r=r_collapse</code> .	C.1; tables/equilibrium_controls.csv	decimal_6
base_frozen_entry_weak	E with fixed full orders and optimal preparation at this strength; not an equilibrium assertion for the investor. Select: <code>experiment=frozen; noise=Laplace; cost_law=atoms; r=r_weak</code> .	C.1; tables/equilibrium_controls.csv	decimal_6

Key	Definition and row selection	Exercise / output	Display
base_hidden_entry_weak	E in the reoptimized price-hidden economy. Select: experiment=price_hidden; noise=Laplace; cost_law=atoms; r=r_weak.	C.1; tables/equilibrium_controls.csv	decimal_6
base_ownership_weak	$O_H = \bar{e}_H/2$, unconditional probability of high-quality challenger ownership. Select: experiment=feedback; noise=Laplace; cost_law=atoms; r=r_weak.	C.1; tables/equilibrium_controls.csv	decimal_6
base_spread_weak	$\Delta_T = t_H - t_L$ from independently checked auction expectations. Select: r=r_weak.	C.1; tables/auction_primitives.csv	decimal_6
base_profit_prior_weak	$B_r(1/2) = (g_H + g_L)/2$. Select: r=r_weak.	C.1; tables/auction_primitives.csv	decimal_6
base_frozen_entry_strong	E with fixed full orders and optimal preparation at this strength; not an equilibrium assertion for the investor. Select: experiment=frozen; noise=Laplace; cost_law=atoms; r=r_strong.	C.1; tables/equilibrium_controls.csv	decimal_6
base_hidden_entry_strong	E in the reoptimized price-hidden economy. Select: experiment=price_hidden; noise=Laplace; cost_law=atoms; r=r_strong.	C.1; tables/equilibrium_controls.csv	decimal_6
base_ownership_strong	$O_H = \bar{e}_H/2$, unconditional probability of high-quality challenger ownership. Select: experiment=feedback; noise=Laplace; cost_law=atoms; r=r_strong.	C.1; tables/equilibrium_controls.csv	decimal_6
base_spread_strong	$\Delta_T = t_H - t_L$ from independently checked auction expectations. Select: r=r_strong.	C.1; tables/auction_primitives.csv	decimal_6
base_profit_prior_strong	$B_r(1/2) = (g_H + g_L)/2$. Select: r=r_strong.	C.1; tables/auction_primitives.csv	decimal_6
base_revenue_feedback	$\mathcal{R}_T$ in the strong feedback equilibrium. Select: experiment=feedback; r=r_strong; noise=Laplace; cost_law=atoms.	C.1; tables/equilibrium_controls.csv	decimal_6

Key	Definition and row selection	Exercise / output	Display
base_revenue_hidden	$\mathcal{R}_T$ in the strong price-hidden equilibrium. Select: <code>experiment=price_hidden; r=r_strong; noise=Laplace; cost_law=atoms.</code>	C.1; tables/equilibrium_controls.csv	decimal_6
base_matched_dividend	$D_0 = \mathcal{R}_T^{feedback} - \mathcal{R}_T^{hidden}$ at the strong strength; verify residual invariance. Select: <code>experiment=matched_dividend; r=r_strong; noise=Laplace; cost_law=atoms.</code>	C.1; tables/equilibrium_controls.csv	decimal_6

Distributional and moderate-value illustrations.

Key	Definition and row selection	Exercise / output	Display
logistic_entry_strong	E under logistic noise and atomic costs. Select: <code>noise=logistic; cost_law=atoms; r=r_strong.</code>	C.1/C.5; tables/equilibrium_controls.csv	decimal_6
logistic_flow_threshold	$x_{\log}^*$ from the likelihood-ratio inverse, independently checked by a root. Select: <code>noise=logistic; cost_law=atoms; r=r_strong.</code>	C.1/C.5; tables/equilibrium_controls.csv	decimal_9
logistic_threshold_noise_sd	$x_{\log}^* / (b\pi/\sqrt{3})$, using the standard deviation of noise, not aggregate flow. Select: <code>noise=logistic; tau=benchmark_strong_threshold; column=threshold_noise_sd.</code>	C.1/C.5; figures_data/posterior_tails.csv	decimal_6
cost_mix_laplace_entry_strong	E under Laplace noise with the complete uniform-mixture cost CDF. Select: <code>noise=Laplace; cost_law=uniform_mixture; r=r_strong.</code>	C.1/C.5; tables/equilibrium_controls.csv	decimal_6
cost_mix_logistic_entry_strong	E under logistic noise with the complete uniform-mixture cost CDF. Select: <code>noise=logistic; cost_law=uniform_mixture; r=r_strong.</code>	C.1/C.5; tables/equilibrium_controls.csv	decimal_6
moderate_entry_weak	E in the moderate-value feedback equilibrium after every strict theorem margin is verified. Select: <code>declared_moderate_comparison; column=E_weak.</code>	C.4; numerics/moderate_values.csv	decimal_6

Key	Definition and row selection	Exercise / output	Display
moderate_entry_strong	E in the moderate-value feedback equilibrium after every strict theorem margin is verified. Select: declared moderate comparison; column=E_strong.	C.4; numerics/moderate_values.csv	decimal_6

Complementary private information.

Key	Definition and row selection	Exercise / output	Display
signal_entry_weak	$E = (\bar{e}_H + \bar{e}_L)/2$, using true-state-conditioned private-signal probabilities. Select: a=0.70; d=0.75; r=r_weak.	C.3; numerics/two_signals.csv	decimal_6
signal_entry_strong	$E = (\bar{e}_H + \bar{e}_L)/2$, using true-state-conditioned private-signal probabilities. Select: a=0.70; d=0.75; r=r_strong.	C.3; numerics/two_signals.csv	decimal_6
signal_trader_accuracy	The input a displayed as a percentage; no new calculation beyond the change of units. Select: a.	C.3; input manifest	percent_integer
signal_buyer_accuracy	The input d displayed as a percentage; no new calculation beyond the change of units. Select: d.	C.3; input manifest	percent_integer

Certified asymmetric equilibria.

Key	Definition and row selection	Exercise / output	Display
cert_a_v_interval	Certified bracket for the exact equilibrium root v^* ; require both endpoint signs, threshold ordering, low-type concavity, and positive global high-type derivative cover. Select: r=1.55; all certificate predicates=true.	C.2; numerics/certificates.csv	outward_interval_8
cert_a_entry_interval	Outward enclosure of $E(r, v)$ for every v in the certified root bracket. Select: r=1.55; all certificate predicates=true.	C.2; numerics/certificates.csv	outward_interval_10

Key	Definition and row selection	Exercise / output	Display
cert_a_high_derivative_lower	The nonnegative lower enclosure Γ_H after subtracting the between-grid Lipschitz correction, uniform over the full root bracket. Select: <code>r=1.55; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	lower_bound_10
cert_b_v_interval	Certified bracket for the exact equilibrium root v^* ; require both endpoint signs, threshold ordering, low-type concavity, and positive global high-type derivative cover. Select: <code>r=1.60; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	outward_interval_8
cert_b_entry_interval	Outward enclosure of $E(r, v)$ for every v in the certified root bracket. Select: <code>r=1.60; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	outward_interval_10
cert_b_high_derivative_lower	The nonnegative lower enclosure Γ_H after subtracting the between-grid Lipschitz correction, uniform over the full root bracket. Select: <code>r=1.60; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	lower_bound_10
cert_c_v_interval	Certified bracket for the exact equilibrium root v^* ; require both endpoint signs, threshold ordering, low-type concavity, and positive global high-type derivative cover. Select: <code>r=1.65; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	outward_interval_8
cert_c_entry_interval	Outward enclosure of $E(r, v)$ for every v in the certified root bracket. Select: <code>r=1.65; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	outward_interval_10
cert_c_high_derivative_lower	The nonnegative lower enclosure Γ_H after subtracting the between-grid Lipschitz correction, uniform over the full root bracket. Select: <code>r=1.65; all certificate predicates=true</code> .	C.2; numerics/certificate.csv	lower_bound_10

Atomless-value reserve comparison.

Key	Definition and row selection	Exercise / output	Display
value_entry_weak_high_p	Class-economy E at the high reserve, with class-only investor information. Select: value_law=uniform_classes; epsilon_V=0.05; r=r_weak; p=1.1.	C.6; tables/reserve_comp arisons.csv	decimal_6
value_revenue_weak_low_p	Class-economy $\mathcal{R}_T$ using integrated within-class auction payoffs and the validated continuation. Select: value_law=uniform_classes; epsilon_V=0.05; r=r_weak; p=0.5.	C.6; tables/reserve_comp arisons.csv	decimal_6
value_revenue_weak_high_p	Class-economy $\mathcal{R}_T$ using integrated within-class auction payoffs and the validated continuation. Select: value_law=uniform_classes; epsilon_V=0.05; r=r_weak; p=1.1.	C.6; tables/reserve_comp arisons.csv	decimal_6
value_entry_strong_high_p	Class-economy E at the high reserve, with class-only investor information. Select: value_law=uniform_classes; epsilon_V=0.05; r=r_strong; p=1.1.	C.6; tables/reserve_comp arisons.csv	decimal_6
value_revenue_strong_low_p	Class-economy $\mathcal{R}_T$ using integrated within-class auction payoffs and the validated continuation. Select: value_law=uniform_classes; epsilon_V=0.05; r=r_strong; p=0.5.	C.6; tables/reserve_comp arisons.csv	decimal_6
value_revenue_strong_high_p	Class-economy $\mathcal{R}_T$ using integrated within-class auction payoffs and the validated continuation. Select: value_law=uniform_classes; epsilon_V=0.05; r=r_strong; p=1.1.	C.6; tables/reserve_comp arisons.csv	decimal_6

Validation and construction scalars. These quantities are produced before figures or replacement of manuscript placeholders. They make the proof-region checks and threshold distinctions machine-checkable even when the corresponding value is not printed in the main text.

Key	Definition	Exercise / output	Display
base_m	$m = (1 + e^{2/b})^{-1}$.	C.2; numerics/thresholds .csv	decimal_9
base_M	$M = 1 - m$.	C.2; numerics/thresholds .csv	decimal_9
base_r_pool_unique_sufficient	$r(k)$, the sufficient pooling-uniqueness boundary.	C.2; numerics/thresholds .csv	decimal_9
base_r_no_trade_exact	$r_N = r(2k/\rho)$, checked within the stated prior/floor domain.	C.2; numerics/thresholds .csv	decimal_9
base_r_full_unique_sufficient	$r_U = r(k/[(1 - 1/b)\rho m])$.	C.2; numerics/thresholds .csv	decimal_9
base_r_high_cost_ceiling	r_C from (OA.20), independently checked against $B_r(M) = c_H$ and its admissible domain.	C.2; numerics/thresholds .csv	decimal_9
base_laplace_entry_ceiling_left_limit	$\rho + (1 - \rho)(1 + e^{-2/b})/4$; the one-sided limit, not automatically the boundary equilibrium.	C.2; numerics/thresholds .csv	decimal_9
base_margin_low_cost	ζ_L from (OA.68); a strict inequality requires a strictly positive verified value.	C.1; tables/extensions.csv	scientific_10
base_margin_high_prior	ζ_{H0} from (OA.68); a strict inequality requires a strictly positive verified value.	C.1; tables/extensions.csv	scientific_10
base_margin_high_ceiling	ζ_{H1} from (OA.68); a strict inequality requires a strictly positive verified value.	C.1; tables/extensions.csv	scientific_10
base_margin_weak_trade	ζ_0 from (OA.68); a strict inequality requires a strictly positive verified value.	C.1; tables/extensions.csv	scientific_10
base_margin_strong_trade	ζ_1 from (OA.68); a strict inequality requires a strictly positive verified value.	C.1; tables/extensions.csv	scientific_10
moderate_margin_low_cost	ζ_L from (OA.68); a strict inequality requires a strictly positive verified value.	C.4; numerics/moderate_values.csv	scientific_10
moderate_margin_high_prior	ζ_{H0} from (OA.68); a strict inequality requires a strictly positive verified value.	C.4; numerics/moderate_values.csv	scientific_10
moderate_margin_high_ceiling	ζ_{H1} from (OA.68); a strict inequality requires a strictly positive verified value.	C.4; numerics/moderate_values.csv	scientific_10
moderate_margin_weak_trade	ζ_0 from (OA.68); a strict inequality requires a strictly positive verified value.	C.4; numerics/moderate_values.csv	scientific_10
moderate_margin_strong_trade	ζ_1 from (OA.68); a strict inequality requires a strictly positive verified value.	C.4; numerics/moderate_values.csv	scientific_10

Key	Definition	Exercise / output	Display
signal_margin_low_cost	ζ_L as defined by the corresponding low-cost, high-cost, or trading inequality in (OA.29); use the signal-specific bounds, not the benchmark bound.	C.3; numerics/two_signal s.csv	scientific_10
signal_margin_high_prior	ζ_{H0} as defined by the corresponding low-cost, high-cost, or trading inequality in (OA.29); use the signal-specific bounds, not the benchmark bound.	C.3; numerics/two_signal s.csv	scientific_10
signal_margin_high_ceiling	ζ_{H1} as defined by the corresponding low-cost, high-cost, or trading inequality in (OA.29); use the signal-specific bounds, not the benchmark bound.	C.3; numerics/two_signal s.csv	scientific_10
signal_margin_weak_trade	ζ_0 as defined by the corresponding low-cost, high-cost, or trading inequality in (OA.29); use the signal-specific bounds, not the benchmark bound.	C.3; numerics/two_signal s.csv	scientific_10
signal_margin_strong_trade	ζ_1 as defined by the corresponding low-cost, high-cost, or trading inequality in (OA.29); use the signal-specific bounds, not the benchmark bound.	C.3; numerics/two_signal s.csv	scientific_10
base_net_surplus_gain	ΔW at strong strength: (OA.47) and the independent difference of (OA.49).	C.1; numerics/feedback_comparisons.csv	decimal_6
base_revenue_gain	$\Delta \mathcal{R}_T$ at strong strength, calculated from conditional entry and independently from mean prices.	C.1; numerics/feedback_comparisons.csv	decimal_6
base_minimum_theorem_margin	Minimum of the five explicitly defined strict margins, retaining the separate component values in the output.	C.1/C.3/C.4; numerics/quantity_registry.csv	scientific_10
moderate_minimum_theorem_margin	Minimum of the five explicitly defined strict margins, retaining the separate component values in the output.	C.1/C.3/C.4; numerics/quantity_registry.csv	scientific_10

Key	Definition	Exercise / output	Display
signal_minimum_theorem_margin	Minimum of the five explicitly defined strict margins, retaining the separate component values in the output.	C.1/C.3/C.4; numerics/quantity_registry.csv	scientific_10

The minimum-margin statistic is a convenience for acceptance, not a substitute for recording every inequality. The threshold values answer different questions; a renderer cannot rename the sufficient full-order uniqueness boundary as an equilibrium activation threshold. For the certified nodes, the source also records the root-endpoint derivative enclosures and the global high-type derivative enclosure even when only their lower bound appears in the manuscript.

Root-sign quantities accompanying the paper-appendix certificates. The root bracket for each node is the one defined by its corresponding interval key. The left lower bound must be positive and the right upper bound negative before either value is substituted.

Key	Definition	Source and row	Display
cert_a_psi_left_lower	Outward endpoint enclosure $\inf \Psi(r, v_-)$.	C.2; numerics/certificates.csv; r=1.55; Psi_left_lower	lower_bound_12
cert_a_psi_right_upper	Outward endpoint enclosure $\sup \Psi(r, v_+)$.	C.2; numerics/certificates.csv; r=1.55; Psi_right_upper	upper_bound_12
cert_b_psi_left_lower	Outward endpoint enclosure $\inf \Psi(r, v_-)$.	C.2; numerics/certificates.csv; r=1.60; Psi_left_lower	lower_bound_12
cert_b_psi_right_upper	Outward endpoint enclosure $\sup \Psi(r, v_+)$.	C.2; numerics/certificates.csv; r=1.60; Psi_right_upper	upper_bound_12
cert_c_psi_left_lower	Outward endpoint enclosure $\inf \Psi(r, v_-)$.	C.2; numerics/certificates.csv; r=1.65; Psi_left_lower	lower_bound_12
cert_c_psi_right_upper	Outward endpoint enclosure $\sup \Psi(r, v_+)$.	C.2; numerics/certificates.csv; r=1.65; Psi_right_upper	upper_bound_12

lower_bound_12 rounds downward and upper_bound_12 rounds upward to twelve decimal places. A bound whose displayed sign becomes zero is printed with additional outward-rounded digits, not rounded inward to preserve a sign.

First production pass. Produce the benchmark, moderate-value, and complementary-signal input rows; the complete margin rows and minimum margins; the benchmark pooling/full-order/expensive-entry boundaries; the benchmark entry, hidden-information, frozen-profile, ownership, and welfare quantities; and the certified asymmetric brackets and derivative bounds. These determine which theorem regions and outcome comparisons can be populated. The exploratory correspondence and reserve sweeps follow them. A figure is not used as a source of a scalar, and an apparent curve is not used to fill a missing certificate.

D. Institutional pilot and empirical design

The empirical part of this project starts with an institutional question rather than a regression. Was there an interval during which a prospective buyer could still decide whether to investigate a publicly visible sale while the target's shares traded? This appendix sets out the rule for finding such intervals, the coding of events, and what would count as evidence that the institution the model describes exists.

D.1. The decision interval and sample-selection rule

The pilot establishes the institution required by the model before attempting to estimate its causal mechanism. The unit of observation is a prospective buyer's participation decision within a publicly visible sale process for a listed target. A process is eligible when contemporaneous public information identifies a plausible acquisition opportunity, the target continues to trade, and at least one economically meaningful participation decision by a potential challenger remains open. Public visibility and the buyer's later decision must be separately dated.

A public strategic review, disclosed approach, or open bidding contest is a candidate starting point, not automatic inclusion. The public record must show that a prospective buyer could still decide whether to investigate or submit a substantive proposal. A transaction announced only after the buyer set was fixed does not qualify merely because the target traded during its earlier confidential negotiations. Likewise, a named firm mentioned by an adviser is a potential buyer, not an entrant; an NDA alone need not establish costly acquisition preparation. I retain each of these events but distinguish their economic content.

Selection proceeds without conditioning on whether a later bidder wins, whether a target return has a particular sign, or whether the process appears to support the theory. Construct a process-level screening log from the eligible filing and announcement universe, record inclusion and exclusion decisions and their evidence, and retain uncertain cases for adjudication. Multiple processes for the same target receive separate process identifiers when the record documents a termination and restart. No cases are selected and no sample count or estimate is reported in this design.

D.2. Information available at the time, not only at the filing date

I distinguish the date an event occurred from the date its content first became public. A definitive proxy can describe a private meeting retrospectively. That description supports the occurrence of the meeting, but does not put it in the stock market's information set on the meeting date. For every economically relevant event, record the occurrence date or bounded date range, the first verified disclosure date, the public source, and the specificity of the disclosed information.

The candidate modeled interval begins when the opportunity and relevant competitive threat are publicly interpretable and ends at the challenger's material participation decision. If the rival's identity, financing position, or proposed terms become public later, those facts enter the information set only then. An unverified rumor is coded as a rumor with its source and uncertainty, not as an established public bid. If the chronology provides only a month or a relative ordering, preserve that precision rather than inventing a daily date.

A process can contain more than one candidate interval. The pilot retains the chronology and identifies each interval explicitly; it does not select the interval producing the strongest return–entry association. Confidential diligence by the challenger may follow public visibility even when its identity remains private to

the market. In that case the record must support an interpretation of what match-related information a public investor could hold, rather than presuming that the market knew the eventual entrant.

D.3. Sources and event coding

The primary chronology is the target’s merger-background discussion in definitive proxy or tender-offer filings, supplemented by current-report filings, transaction exhibits, contemporaneous company releases, and time-stamped public reporting. SEC accession numbers and exact source locations accompany every coded factual claim. A filing’s narrative is evidence, not an instruction to treat every party as having the same information. Contradictory dates or actor descriptions remain visible until resolved.

I use the following event categories in the pilot. An initial approach identifies who initiated contact and whether it was solicited. Public visibility identifies the first disclosure of the acquisition opportunity and the competitive facts revealed. Buyer contact records outreach and replies without equating them with entry. Confidentiality and data access record when information became available to the buyer. Diligence entry records a substantiated commitment to evaluation, including its stated scope. A first substantive proposal records price, consideration, financing, and diligence conditions. A revised proposal records what changed and which new information preceded it. Withdrawal records the actor making the decision and the reason when stated. Final selection records the board’s choice, agreement, or process termination. Announced and actual deadlines are separate events.

The event ledger has a row for each event–actor link. The following schema is a data specification, not an extracted dataset:

pilot/processes.csv:

```
process_id, target_id, target_name, listing_status,  
process_start_date_lower, process_start_date_upper,  
public_visibility_date_lower, public_visibility_date_upper,  
process_end_date_lower, process_end_date_upper,  
lead_buyer_id, process_status, eligible, exclusion_reason,  
information_interval_established, unresolved_issue
```

pilot/events.csv:

```
process_id, event_id, actor_id, actor_role, event_type,  
event_date_lower, event_date_upper, event_date_precision,  
first_public_date_lower, first_public_date_upper,  
public_information_content, contemporaneous_or_retrospective,  
price_or_range, consideration, financing_condition,  
diligence_condition, binding_status, initiation_actor,  
withdrawal_decision_actor, reported_reason,  
accession_or_source_id, source_locator, verbatim_evidence,  
source_publication_time, evidence_confidence, adjudication_status
```

pilot/intervals.csv:

```
process_id, interval_id, challenger_id,
```

interval_start_lower, interval_start_upper,
participation_decision_lower, participation_decision_upper,
public_opportunity_evidence, public_incumbent_evidence,
participation_evidence, information_complementarity_interpretation,
ordering_verified, eligibility_decision, unresolved_issue

pilot/adjudication.csv:

record_id, field, initial_value, alternative_value,
conflict_type, supporting_sources, resolution, resolver,
resolution_date, unresolved_reason

An absent field means missing, not applicable, or not stated, with a corresponding reason; it is not a zero price, an absence of a buyer, or a voluntary withdrawal. For price ranges, store both endpoints rather than the midpoint alone. A proposal's classification as formal or informal follows the source's description and the coded contractual features; no uniform price threshold or arbitrary document label is used to manufacture a distinction across processes. Quotations are checked against the source location, and summaries are stored separately from the evidence text.

D.4. What would establish institutional relevance

The pilot supports the timing if it documents a public opportunity, trading during that opportunity, and a subsequent participation decision that remained genuinely open. It supports the preparation margin if a buyer had to commit resources or obtain additional target information before submitting an executable bid. It supports an incremental-information interpretation if the relevant uncertainty was not already fully resolved by the buyer's public statements and the record leaves room for distinct target-side and buyer-side knowledge.

None of these observations alone establishes that prices caused entry. A target price rising before a competing bid can anticipate that bid. A private process described after announcement does not show that the public market knew it beforehand. A buyer's later success does not prove that it entered after receiving a favorable price signal. Such cases may illuminate institutions while remaining outside the mechanism's direct empirical evidence.

The first output is a sourced chronology and an accounting of eligible decision intervals, with exclusions and ambiguous cases retained. The next output is a short institutional section describing which elements of the modeled game occur in the verified intervals. The existing process illustration in the manuscript is not a substitute for this selection and coding exercise.

D.5. Measurement and eventual empirical tests

The participation outcome is commitment to investigation or substantive bidding, measured at the most precise date the evidence supports. Public bid counts are a separate, narrower outcome. High-quality ownership in the theory is a structural value concept; a winning challenger is observable, but winning alone is not evidence that its underlying acquisition value was high. Any empirical proxy for match quality must be separately justified.

Incumbent strength is measured using information available before the challenger's decision. A final

offer reflects subsequent entry and cannot serve as a predetermined initial threat. Candidate descriptive measures include an initial disclosed proposal, pre-existing financing capacity, and publicly measurable buyer–target complementarity. Their interpretation must respect whether the measure changes competitive strength, common acquisition value, financing constraints, or several objects at once.

Returns, trading activity, and other information measures are timestamped before the outcome. Define the public-information cutoff before constructing financial windows, and retain alternative date bounds when the event time is interval-censored. A stock return is not itself a measure of how much information was revealed. The empirical design must distinguish a change in the expected level of sale proceeds from a change in the market’s information about challenger quality.

A targeted causal design would shift information transmission without independently changing acquisition values, preparation costs, or financing. No such instrument is asserted here. Alternatively, a structural exercise could jointly model anticipated entry and learning from prices, using the final institution-specific theorem to identify discriminating restrictions. The pilot determines which route is institutionally credible before either exercise is undertaken.

E. Reproducibility and the numerical boundary

This appendix records how the numbers in the paper were produced and where the computational boundary lies. I distinguish the node checks distributed with the paper from the full exercises of Appendix C, and I state the environment in which each was run.

E.1. Distributed verification and execution environment

The verification directory contains independent node calculations, exploratory asymmetric searches, and interval certificates. The documented execution environment is Python 3.13.5, NumPy 2.3.5, SciPy 1.17.0, and mpmath 1.3.0. These are metadata for the distributed results, not a claim that every other compatible environment fails. Each new execution records its actual interpreter, package versions, operating system, arithmetic precision, parameter declarations, tolerances, and source-file hashes.

The computational contract in Appendix C extends beyond the distributed node checks. In particular, the complete strength correspondence, the exploratory reserve correspondence, and the figure-ready scalar registry are exercises to execute and validate. Their outputs are not inferred from the existence of a node verifier or filled from undocumented cached calculations. The empirical pilot contains a coding specification rather than an extracted case sample.

To reproduce the distributed checks from the verification directory, use:

```
python -m pip install -r requirements.txt
python review_checks.py --out results --mode core
python review_checks.py --out results --mode signals
python explore_asymmetric.py
python certify_asymmetric.py
```

Run in an isolated working copy and retain the distributed result files separately from newly generated results. Record a failure as a failure; do not change a tolerance or catch an exception solely to obtain an accepted output. The interval certificate requires outward interval arithmetic and its analytical cover.

Replacing it with ordinary floating-point quadrature changes the evidentiary status.

E.2. Contents and scope of the distributed files

The core verifier computes auction expectations, theorem margins, full-order objects, threshold distinctions, moderate values, noise-law comparisons, and the specified class-value reserve alternatives. Its signal mode uses the conditional laws of Appendix A.7. The exploratory asymmetric search identifies candidate roots and checks numerical deviations. The certificate evaluator uses the antiderivatives and derivative cover of Appendix B to establish selected continuous-action equilibria.

The distributed directory contains:

```
verification/  
  review_checks.py  
  explore_asymmetric.py  
  certify_asymmetric.py  
  requirements.txt  
results/  
  core_results.json  
  deviations.csv  
  two_signal_results.json  
  two_signal_deviations.csv  
  asymmetric_candidates.json  
  asymmetric_interval_certificates.json  
  certificate_run_log.txt
```

The core results hold primitive vectors, inequality margins, threshold calculations, candidate outcomes, and reserve comparisons. The deviation ledger records tested state–order alternatives and their payoffs. The signal results hold private-information margins, joint-probability checks, and the associated deviation ledger. The asymmetric candidate file is exploratory: its root and local-search output is not an exhaustive equilibrium enumeration. The interval file contains the root-sign enclosures, order-cover bounds, and entry intervals that complete the selected computer-assisted existence arguments. The certificate log provides the readable arithmetic output; the complete interval endpoints, not their displayed midpoints, are the numerical proof objects.

These files support the statements at their documented parameter points. They do not supply an optimizer over sale terms, a first-price bidding equilibrium, or the full set of mixed trading equilibria. The corresponding research questions are implemented through the new exercises in Appendix C rather than relabeled node output.

E.3. Implementation contract for the full exercises

Use pure numerical functions with explicit parameter arguments and immutable returned records. Separate the auction-payoff layer, information and price construction, unilateral-deviation evaluation, equilibrium search, independent validation, and figure/table rendering. A search routine returns candidate equilibria and unresolved nodes; only the independent validation layer assigns an accepted label. No renderer solves an

equilibrium, silently changes a parameter, or drops an inconvenient branch.

The only boundary to the presentation layer is the validated CSV output specified in Appendix C. Join tables using the complete parameter declaration and branch label, never only a rounded strength or price. Assemble the extensions table from the distributional, moderate-value, and signal outputs, retaining fields that are not applicable as explicitly not applicable. Do not fill a missing signal bound or an uncomputed welfare statistic with zero. The reserve table uses both its declared comparisons and the separately identified found-continuation ranges.

The scalar registry is generated from validated rows and exact input declarations. The substitution stage checks that every placeholder key is defined once, that its source row exists and is unique under the declared selector, and that its result status permits the intended statement. It fails before creating a filled manuscript if a required value remains open. Certificates are substituted as outward intervals and conservative lower bounds. Ordinary diagnostics carry their specified rounding without being promoted to certified enclosures.

Every completed exercise exports a manifest with inputs, method, tolerances, output hashes, and pass/fail outcomes. Failed searches and arithmetic retries remain in the record. This permits a reader to distinguish a genuinely rejected candidate from a numerical failure, and an accepted node from an exhaustive characterization.

E.4. Figure and table rendering

Figures have no internal titles. Captions state the economic comparison, parameter specification, information regime, units, and result-status interpretation. Multi-panel figures use the panel labels specified in the manuscript. The correspondence figure shows separate branches and breaks lines at unresolved nodes. Certified nodes retain interval bars even where the intervals are visually small. A uniqueness boundary is labeled by the theorem it satisfies, not by the first root found by a solver.

Tables preserve the difference between equilibrium comparisons and fixed-profile controls. A frozen-profile row can contain profitable investor deviations without contradicting its purpose; it is not labeled an equilibrium. Welfare columns distinguish target revenue from allocation surplus net of preparation costs. Reserve ranges are described as ranges across continuations found unless an exhaustive argument establishes more. No selected branch, matched mean price, or apparent hump replaces the underlying equilibrium checks.

The completed numerical layer therefore produces interpretable economic objects and an auditable validation record together. It does not turn a partial search into a theorem by hiding the search's scope.